

Drivers of interest rate dispersion in credit markets

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ABSTRACT

Using 1mln observations at the loan-, bank-, firm- level in France, and public rating scores, I document that interest rate differentials that are not explained by observables at loan issuance contain forward information about changes in firms' default probability. I find that this correlation is heterogeneous in the cross section: it is stronger in ex-ante safer rating classes, stronger for young, small firms, and during crises. This evidence overall suggests that banks anticipate information. I propose a model of loan pricing where dispersion is determined by (i) banks' optimal pricing strategy in an imperfectly competitive environment and (ii) banks pricing credit risk differences across firms, depending on available public information. Information plays a major role in determining surplus sharing between banks and heterogeneous firms, and in limiting the consequences of adverse selection. The calibration of the model rationalizes the empirical findings.

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1 Introduction

Credit markets exhibit substantial *interest rate variation across firms and banks*. Most of this variation is accounted for by macroeconomic conditions and observable fundamental factors, including observable credit risk, contract characteristics and bank-firm relationships - see [Altavilla et al. \(2024\)](#). However, a non-negligible share of variation remains unexplained, as documented in many recent papers in different settings and countries: Portugal [Ferreira and Kozeniauskas \(2026\)](#), US [Amiti et al. \(2026\)](#), Italy [Bocola et al. \(2026\)](#), Brazil [Cavalcanti et al. \(2023\)](#). In France, I show that unexplained dispersion after controlling for observables is substantial, more than 15%. The natural question that arises is how much of this dispersion is attributable to unobserved credit risk, versus market inefficiencies that allow deviations from the law of one price.

I address this question empirically, by showing that interest rate residuals contain forward information about changes in firms' credit rating, and that this is in line with bank screening. Then, I propose a model of loan pricing with imperfect competition and imperfect public information about firms. Accounting for both frictions is crucial to understand market complexity and to capture important implications for surplus sharing between banks and heterogeneous firms, as well as how the value banks place on information interacts with competition. Understanding the drivers of unexplained dispersion is also relevant for policy: dispersion arising in an imperfectly competitive environment but unrelated to firms' fundamentals has consequences on firms' investment decisions and the fairness of competition, and overall provides a rationale for regulation and policy intervention. On the other hand, dispersion induced by information on firms' true credit risk reflects the efficiency of the banking market in overcoming asymmetric information, and any policy intervention is at best unnecessary.

I start from characterizing the time and cross-sectional distribution of unexplained rate dispersion by leveraging detailed micro-data on lending in France, spanning from 2006 to 2024. Observable time, bank, firm and loan characteristics account for 83% of total rate variation, which is consistent with the findings in [Altavilla et al. \(2024\)](#) for Euro area loans. The distribution of unexplained rate residuals is substantial (around 1pp), varies over time and firm characteristics, and responds to monetary policy shocks - a tightening reduces total residual variance. The first question is whether unexplained rate residuals may reflect unobservables about the borrowers that banks discover, and that feed into the interest rate. I use firms' evolution in credit quality to document a positive and significant correlation between paying a higher rate and an unexpected increase in future default probability. Exploiting the cross-section, I document that the correlation is stronger where more can be learned about the borrower and where that knowledge matters more for the loan: during the financial crisis, for young and small firms, and

for loans with longer maturity. I directly exclude that higher rates may induce a credit quality deterioration by testing the transmission of interest rate changes to firm-level risk in my sample, exploiting the interest rate structure of the contract, fixed versus floating. Overall, this evidence suggests that observed dispersion is due at least in part to information about borrowers that banks price but that I do not observe, while the rest can be attributable to some degree of imperfect competition.

I move to studying a tractable theoretical framework that determines loan pricing as the outcome of imperfect competition, modelled via search frictions, and of the information the market makes available about firms' true types. The model features two symmetric banks and a continuum of firms. There are search frictions in this market - modelled as in [Burdett and Judd \(1983\)](#):- some firms sample both banks' offers and compare them, others may be stuck with one bank's offer only; banks cannot distinguish captive and shopper firms. There is a second friction in this market: information asymmetry. Firms can be of two types, differing in their riskiness and reservation rate; banks cannot distinguish between types on their own, but the market generates a public, imperfect signal on every firm, which banks can consult at a cost. Two features of this signal shape the analysis. It is *public*: both banks read the same realization, so neither holds an informational advantage over the other, and the asymmetry runs between banks and firms rather than between banks. And its precision q is a *feature of the market* — how much can be learned about a borrower where it operates, given credit registries, reporting histories, disclosure rules and rating coverage — rather than a decision of the banks, who take it as given and choose only whether to pay for it. I characterize the equilibrium dispersed rate distribution, showing that both frictions jointly contribute to observed dispersion. Search frictions deliver *within-type dispersion*. In a non-perfectly competitive environment, banks compete for borrowers, but search frictions prevent them from perfectly undercutting each other; consequently, a mixed-strategy equilibrium arises, and even homogeneous firms face different rates. The information structure of the market affects whether banks pool or separate borrowers, determining *between-type dispersion*; where the market's signal is very precise, good types will pay on average lower rates than bad types, leading to a positive correlation between rates and default probability.

I run a calibration exercise to recover the implied parameters of competition and signal precision from the unexplained rate distribution across Banque de France rating classes. I target 3 data moments: the total variance of rate residuals, decreasing in competition and increasing in signal precision; the correlation between rates and riskiness, increasing in both competition and signal precision; the maximum rate observed. The exercise suggests that the equilibrium is close to perfect information in ex-ante safer markets, with information decreasing as I move to ex-ante riskier rating classes. However, the relative

contribution of competition to unexplained variance is predominant (0-5%), even under perfect information. Because the signal is public, both banks read the same evidence on every firm, so a common precision is a property of the environment rather than an equilibrium selection. The calibration also hides substantial differences in how the surplus is shared across banks and types.

From the theoretical framework, I derive important implications for surplus sharing. Under search frictions only, but perfect information, all firms get credit and the full gains from trade are realized; the stronger the search frictions, the larger the banks' share of total surplus. More interestingly, the information structure has dramatic implications for how much surplus is lost and for how it is split between banks and firms. When adverse selection is severe, good types either cross-subsidize bad types in a pooling equilibrium, or are pushed out of the market altogether, so their gains from trade go unrealized; more information raises total surplus and reduces the subsidy.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Sections 3 and 4 present the data and the stylized facts, including the evidence that rates carry forward information on firms' credit risk. Section 5 outlines the theoretical framework. Section 6 details the calibration and quantitative results. Section 7 analyzes the implications for surplus sharing. Section 8 concludes.

2 Related Literature

This paper connects several strands of literature: the empirics of interest rate dispersion, the theory of price dispersion under imperfect competition, the theory of adverse selection and screening, and the work on information acquisition and market power in banking.

Rate dispersion in loan markets. In credit markets, rate dispersion after controlling for observable fundamentals is the object of many papers leveraging loan-level data in different contexts. [Cavalcanti et al. \(2023\)](#) document dispersion across observationally equivalent firms in Brazil and focus on the consequences on capital misallocation and economic development; they show how this mechanism can contribute to differences in economic development across countries. [Altavilla et al. \(2024\)](#) decompose the variation in rates on Euro Area loans into macroeconomic fundamentals, bank identity, firm and loan characteristics, which overall account for 80–85% of total rate variation. I obtain a similar share of explained variation, but I focus on the unexplained component of loan pricing. Using French data, [Mazet-Sonilhac \(2025\)](#) provides evidence on the existence of search frictions in the French loan market, which motivates a framework where search and information frictions interact. I engage in the ambitious exercise of disentangling the

relative contribution of both frictions to observed dispersion. [Amiti et al. \(2026\)](#) use US large-loan data to document dispersion within narrow bins of observationally equivalent firms and quarters; measures of banks’ private information about default risk (internal ratings) fail in explaining rate variation, and they conclude that search frictions account for most of the unexplained variation. In my setting, lacking direct measures of banks’ private information, I will find an innovative way to extract the information component: I use the future evolution of firms’ credit risk, as summarized by external credit ratings. [Ferreira and Kozeniauskas \(2026\)](#) similarly documents “idiosyncratic noise” in financing rates arising from heterogeneous search abilities, with default risk explaining little of the residual variation. In line with these papers, I find that search frictions play a predominant role in explaining rate dispersion, calling for policy intervention. However, I depart from these papers by showing that the low contribution of banks’ private information about default risk to rate dispersion does not imply that banks are not screening at all. On the contrary, I will find that banks have strong screening incentives, particularly in buckets of firms with ex-ante low risk; omitting this component can mislead the understanding of the role of screening in loan pricing, and leads to overstating the contribution of search frictions to observed dispersion.

Price dispersion in mortgage markets. A closely related body of evidence comes from mortgage markets, where residual dispersion has been traced to price discrimination, borrower riskiness and switching costs. [Woodward and Hall \(2012\)](#) and [Bhutta et al. \(2021\)](#) attribute a large share of it to shopping intensity: borrowers who solicit few quotes, or who are less financially sophisticated, systematically overpay for otherwise comparable loans, which is the household-market counterpart of the search friction I model. [Coen et al. \(2024\)](#) document price dispersion in the UK mortgage market and show that the price differences they measure are not driven by borrower risk, as gauged by realised default outcomes, but by heterogeneity in the menus of offers that lenders construct: lenders price-discriminate against borrowers who are likely to choose poorly, while competing for those who shop more effectively. Menu heterogeneity is therefore a third source of dispersion, distinct from search and from screening. I believe I can abstract from it for two reasons. First, I compare loans that are similar in both firm and contract characteristics, assuming that the rate varies linearly in the characteristics I control for, so that differences in contract design are absorbed rather than left in the residual. Second, within a given lender–month the loan terms I observe follow rather similar characteristics, which limits the scope for menu-based discrimination within my comparison groups. In line with this literature, I find that search frictions play an important role in explaining dispersion; I depart from it in also finding evidence of an unobserved risk component in dispersed rates, which I recover from the forward information contained in rate residuals.

Price dispersion as an equilibrium outcome. My model builds on the literature showing that identical sellers can charge different prices in equilibrium. [Diamond \(1971\)](#) establishes the benchmark result that this literature has to overturn: if all sellers are identical and all buyers pay the same small search cost, the only equilibrium is the monopoly price. Dispersion therefore requires that buyers differ in what they observe. [Varian \(1980\)](#) shows that when some consumers know all prices and the others know only one, sellers have no pure-strategy best response: cutting the price wins the informed buyers, raising it exploits the uninformed ones. Sellers then randomize, and prices differ even though costs and quality do not. [Burdett and Judd \(1983\)](#) obtain the same result when buyers differ in how many quotes they sample, and [Stahl \(1989\)](#) shows that dispersion also arises under sequential search, with monopoly pricing and competitive pricing emerging as limiting cases when the share of shoppers goes to zero or to one. [Baye et al. \(1992\)](#) describe the full set of mixed-strategy equilibria in this class of models.

I adopt this structure to explain why banks quote different rates to firms that are observationally equivalent, but my environment departs from it in two respects. First, the seller's cost is buyer-specific and unobserved: the return on a given quote depends on the borrower's hidden type, so the equilibrium rate distribution has to reconcile banks' indifference over the rates in the support with the participation constraints of both types. Since the two types differ in their reservation rate, the support is partitioned at those thresholds, and whether the lower segment is offered to both types depends on the severity of adverse selection, which governs the existence of a pooling region. Dispersion is therefore not only within-type, as in the papers above, but also between-type. Second, the precision of banks' information is an equilibrium object rather than a primitive: banks pay for a signal, and its value is governed by the profits they expect to earn, which in turn depend on the intensity of competition. The two frictions thus shape the rate distribution jointly, and the extent of dispersion responds to the return on screening as much as to the search friction itself. Finally, I bring the resulting distribution to loan-level data and recover its structural parameters, so that the relative weight of the two forces becomes a quantitative question rather than a theoretical possibility.

Adverse selection, screening and the cost of pooling. The second ingredient is the theory of markets where quality is hidden, which delivers between-type dispersion and my results on surplus sharing. [Akerlof \(1970\)](#) shows that asymmetric information can unravel a market. [Rothschild and Stiglitz \(1976\)](#) give the standard screening framework: an uninformed seller can separate types only by distorting the contract offered to the good type, so competitive screening leads to separating contracts, although an equilibrium may not exist. [Stiglitz and Weiss \(1981\)](#) bring this logic to credit markets:

because higher rates worsen borrower selection and reduce expected returns, banks may ration credit rather than increase rates, and under pooling good borrowers subsidize bad ones. [Bester \(1985\)](#) shows that rationing disappears if lenders can screen, in his case by offering menus of collateral and rates. My surplus-sharing results are the quantitative version of this argument. When the signal is imprecise, my model produces the same two costs: good types pay a pooled rate that subsidizes bad types, and when adverse selection is severe their offer exceeds their reservation rate and they leave the market. A more precise signal reduces both. The difference is in how types are separated: not through the contract (collateral, quantity or menus) but through costly information, so separation is a matter of degree and can be estimated from the data. [Lester et al. \(2019\)](#) are the closest link between these literatures, as they add screening through menus to a market with search frictions and adverse selection. I find no evidence of the rate-quantity relation that menus would imply in my sample, so I model screening as information acquisition instead; the implications for surplus sharing and for cross-subsidization from good to bad firms are similar.

Information acquisition and market power in credit markets. *Disentangling search frictions and information* is challenging as they most likely interact in my setting. Banks' private information is a source of market power, and search frictions have consequences on the birth of new lending relationships and banks' screening ability. [Carletti et al. \(2024\)](#) provide a comprehensive review of how market power and informational advantages interact in banking, and [Dell'Ariccia \(2001\)](#) shows that the informational advantage of incumbent banks works as a barrier to entry, so market structure is itself shaped by information frictions. The interaction between competition and information acquisition has been directly theorized in credit markets: [Hauswald and Marquez \(2006\)](#) show that banks invest in information technology to screen borrowers, but stronger competition reduces the return on this investment and leads to under-screening; [Dell'Ariccia and Marquez \(2006\)](#) show that lending booms driven by intensified competition generate looser lending standards, as banks cannot recoup screening costs when rivals can poach identified good borrowers. Both papers provide theoretical grounding for the competition-information trade-off at the heart of my model. On the empirical side, [Petersen and Rajan \(1994\)](#) show that lending relationships improve credit availability and terms for small firms, evidence that banks hold information that is not public; [Claessens et al. \(2025\)](#) use internal bank ratings to isolate the private information component in loan terms, finding that it significantly affects rates, collateral, and maturity; [Agarwal and Hauswald \(2010\)](#) show that geographic proximity to the borrower reduces informational asymmetries and improves pricing accuracy. Central to my interpretation is the *hold-up problem*: [Sharpe \(1990\)](#) shows how relationship lending generates informational rents, as a bank that learns

about its borrower gains an informational advantage that weakens the borrower’s outside option, and [Rajan \(1992\)](#) builds on this to study the borrower’s choice between informed and arm’s-length debt; [Cahn et al. \(2024\)](#) show that the public disclosure of credit ratings attenuates this hold-up by reducing banks’ informational monopoly. [Ioannidou and Ongena \(2010\)](#) document that firms switching banks receive rate discounts, consistent with incumbents extracting rents from informationally captive borrowers. [Flanagan \(2023\)](#) estimates that screening costs are embedded in loan rates and are larger for more opaque borrowers, directly linking information acquisition to pricing. Related sources of market power include *switching costs and relationship rents*: [Drechsler et al. \(2017\)](#) demonstrate that concentrated deposit markets allow banks to maintain wide spreads; [Petersen and Rajan \(1995\)](#) show that greater competition reduces relationship lending;¹ [Degryse and Ongena \(2005\)](#) document spatial price discrimination, with banks charging higher rates to distant borrowers who are less likely to search; and [Boot and Thakor \(2000\)](#) show that intensified competition erodes banks’ incentives to invest in relationship-based screening. The paper most closely related to mine is [Crawford et al. \(2018\)](#), who jointly estimate asymmetric information and lender market power in Italian mortgages and find both frictions quantitatively important; I differ in two ways: I model active information acquisition by banks rather than screening via menus, as I do not find a positive correlation between loan take-up and rates in my data; second, I also find important interactions, but while they find that market power can mitigate the effects of adverse selection (prices increase by less when banks have market power and adverse selection worsens), I find that market power gives banks enough incentive to screen, as they would recover most of the surplus loss induced by adverse selection. In a mortgage market setting, [Agarwal et al. \(2024\)](#) develop a structural model combining search frictions with lender screening, finding that both jointly determine credit allocation and calling for extensions to the business loan market, which this paper pursues. In my setting, I assume for simplicity that search frictions are exogenous to market conditions; however, banks’ screening incentives will depend on the degree of competition in the market via their effect on expected profits.

Search frictions and information in other financial markets. The search friction framework has been widely applied *beyond bank credit*. [Duffie et al. \(2005\)](#) develop the foundational search model for over-the-counter (OTC) financial markets, showing how search costs generate bid-ask spreads and dealer rents. [Farboodi et al. \(2025\)](#) show that intermediaries extract rents precisely because search frictions give them market power over their clients. In the interbank money market, [Bech and Monnet \(2015\)](#) apply a random search model to explain the dispersion of overnight rates and trading volumes in OTC reserve markets. On the theoretical side, [Guerrieri et al. \(2010\)](#) characterize

¹For a survey of relationship lending, see [Boot \(2000\)](#); [Berger and Udell \(1995\)](#) and [Berger and Udell \(2002\)](#) provide foundational empirical evidence on how relationship banking benefits small firms.

equilibrium in competitive search markets with adverse selection, where high-quality sellers separate through the prices they post, generating a positive relationship between prices and quality in equilibrium. This mechanism is closely related to the rate-risk correlation I document in my data, and to the separating equilibria that arise in my model under information acquisition.

Implications for the pass-through of policy rates. Monetary policy is not the focus of this paper, but my decomposition speaks to the literature connecting the structure of the loan market to the pass-through of policy rates. The deposit market power documented by [Drechsler et al. \(2017\)](#) lets banks keep spreads wide when policy rates move; [Scharfstein and Sunderam \(2016\)](#) find that market power in mortgage origination dampens pass-through; and [Wang et al. \(2022\)](#) find that bank market power weakens the transmission to corporate investment. Closer to my setting, [Puglisi \(2023\)](#) documents dispersion in the US auto loan market and shows that it generates asymmetric pass-through across borrowers, and [Delis et al. \(2025\)](#) show that tighter credit conditions fall mostly on weaker borrowers. Since I find that most of the residual dispersion comes from imperfect competition rather than from risk-based pricing, my results suggest that such effects may operate through a large part of the rate distribution; the pooling regions of my model, where banks quote the same rate schedule to both types, would add a further source of rigidity. I leave a full treatment of transmission to future work.² I also contribute to the literature that exploits the interest rate structure to evaluate the effects of monetary policy on firms’ liabilities and outcomes; for example, [Gürkaynak et al. \(2022\)](#) use stock market equity prices to show that investors price-in policy rate changes more when firms have a higher share of floating rate liabilities; [Core et al. \(2025\)](#) also exploit the difference between floating and fixed rate loans to show that the effect of a monetary policy tightening on inflation is weakened if firms facing higher borrowing costs adjust prices

3 Empirics

I have combined multiple administrative datasets on business loans, bank-firm relationships and public credit ratings assigned by the French central bank. The resulting loan-level dataset collects all loan terms, firm financial and demographic information and the evolution of default probability after loan issuance. I use this dataset to compute interest rate residuals after controlling for observables at loan issuance, and I show that they contain forward information about unexpected changes in firms’ default probability each year after loan issuance. This correlation is positive and significant, particularly at shorter horizons. Moreover, it is heterogeneous across market segments, particularly strong for

²The shock series reported in the Appendix (Fig. B.9) is built from the high-frequency Euro Area monetary policy surprises of [Altavilla et al. \(2019\)](#).

young, small firms, during financial crises and at longer maturities. I test whether the direction of causality runs from rates to rating worsening in a quasi-experimental setting. Overall, this evidence suggests that forward-looking information embedded in rates reflects information anticipated by banks at the time of loan issuance.

3.1 Data

Loan level data. The main dataset is a quarterly survey of all new loan contracts and credit lines to businesses in the first month of each quarter from 2006 Q1 to 2024 Q2 (*“Nouveaux Crédits”*, provided by Banque de France). It includes anonymized bank identifiers, at the bank unit level, a wide range of loan terms (rate, amount, maturity, loan purpose, fixed or variable rate, pledged guarantees), firm-level identifiers and information on firms’ demographics (age, sector, location, group affiliation), and financial and credit information (including balance-sheet size, debt structure, credit rating score, short-term due repayments, proportion of drawn and undrawn credit lines, revenues).

Bank-firm relationships. I recover information on bank-firm relationships from the French credit register (*“Service Central des Risques”*, SCR). It collects monthly data on outstanding loans of at least €25,000 and credit lines granted by credit institutions and investment firms to companies. I use it to track the length of the existing relationship at loan issuance, whether the firm has multiple bank relationships, and to compute concentration indexes at the department-quarter level.

Public credit ratings. The dataset *“COTA”* contains the credit rating assigned annually by Banque de France to firms and it represents a forward-looking assessment of creditworthiness over a three-year period. Firms are reviewed when their turnover exceeds €750K, (€1250K from 2024). As shown in Fig. A.1, the rating scale has 12 rating categories, including one entry for liquidation (P) and one for when there is not enough information to assign a rating (0). A firm is evaluated on the basis of: the latest balance-sheet data available; data on default/delay events in payments (CIPE); sector-specific information; qualitative information, including interviews with firm managers on an annual basis, to assess the risks and difficulties the firm is facing. This scale was replaced by to a more granular one from January 2022; to ensure continuity, I use Banque de France’s conversion table to convert the new scale to the old one.

Cleaning and sample. I focus on fixed-term loan products to private companies, so I exclude credit lines/other types of credit with undefined maturity; I exclude syndicated and regulated loans. I also recover whether a firm is granted multiple loans or other products in the same month. Finally, I set to missing loan-level observations for which total firm assets fall below the 0.1st or above the 99.9th percentile, and express every

balance-sheet control and the loan-to-assets ratio as a percentage of total assets. After cleaning, the dataset has 1.9 million observations and 0.9 million unique firms. I report a descriptive table of the main loan and firm characteristics for the regression sample, by rating class. The average loan is around €1 million, with substantial heterogeneity between rating classes, and it represents between one fourth and one third of the firm’s total bank debt.

Rating Scale	N obs.	rate (mean)	rate (s.d.)	Quantity (mean, K-Eur)	Maturity (mean, months)	Var. Rate (in %)	Age (mean, years)	Loan/Assets (mean, in %)	Bank Debt/Assets (mean, in %)
3++	7948	2.1	1.8	3754.5	59.0	35	39.0	5.4	19.6
3+	22681	2.2	1.8	2919.3	53.5	37	33.7	5.0	20.6
3	43073	2.3	1.8	2271.5	42.4	37	31.7	5.3	22.0
4+	71629	2.8	2.1	1092.2	34.8	39	28.1	6.2	23.0
4	112974	2.8	2.0	670.2	27.9	43	26.1	7.1	25.0
5+	106418	2.9	2.0	545.5	23.9	48	23.4	7.7	30.3
5	46510	3.5	2.2	781.8	17.9	51	23.1	8.3	31.0
6	13625	3.7	2.2	923.4	16.6	52	21.4	10.4	34.2
7	2657	4.2	2.3	134.7	14.8	62	19.7	8.6	32.7
8	1592	5.0	2.6	107.8	13.2	61	18.0	8.4	31.2
9	268	5.5	2.7	134.0	11.9	48	16.5	10.0	31.7
P (liquidation)	3716	5.0	2.0	114.5	3.4	26	21.8	5.7	33.4
0 (no rating)	24777	3.2	2.1	1895.8	45.0	40	16.1	12.4	27.2
Total	457868	2.9	2.1	1098.1	30.5	44	25.8	7.3	26.5

Table 1: Descriptive table of the loan-level regression sample. Source: Banque de France (Nouveaux Credits, COTA, SCR, FIBEN). Rate: total effective rate (including all fees).

3.2 Interest Rate Residuals

Baseline approach. I measure interest rate residuals by regressing each loan’s interest rates on observables at loan issuance (see [Altavilla et al. \(2024\)](#), [Amiti et al. \(2026\)](#) for the closest approach): I gradually include fixed effects at the macro- (month, lender) and micro- (loan and firm characteristics) levels, and all observed loan and firm characteristics. For each firm-bank-loan-time observation $\text{Rate}_{i,b,l,t}$, I run:

$$r_{ilbt} = \alpha + \mathbf{x}'_{ilbt} \gamma + \mu_{bt\tau s} + \varepsilon_{ilbt}, \quad (1)$$

where time t is quarter, b is bank, i is firm, l is loan. The vector $\mathbf{x}_{ilbt}$ holds what the bank observes at issuance: log loan size, loan over assets, maturity, variable rate dummy, personal loan dummy, individual entrepreneur dummy, firm age and age squared, young firm dummy, new bank-firm relationship dummy, dummies for firms taking several products or several loans in the quarter, and, as a share of assets, short-term debt, turnover, pledged guarantees, undrawn and drawn credit, bank and non bank debt. Fixed effects include $\text{Month} \times \text{Bank} \times \text{Loan type} \times \text{2-digit Sector}$, which captures the supply schedule for similar products for each bank; then I include region, size and rating fixed effects, and a group-membership dummy. The regression sample includes 450 thousand observations and 120 thousand firms.

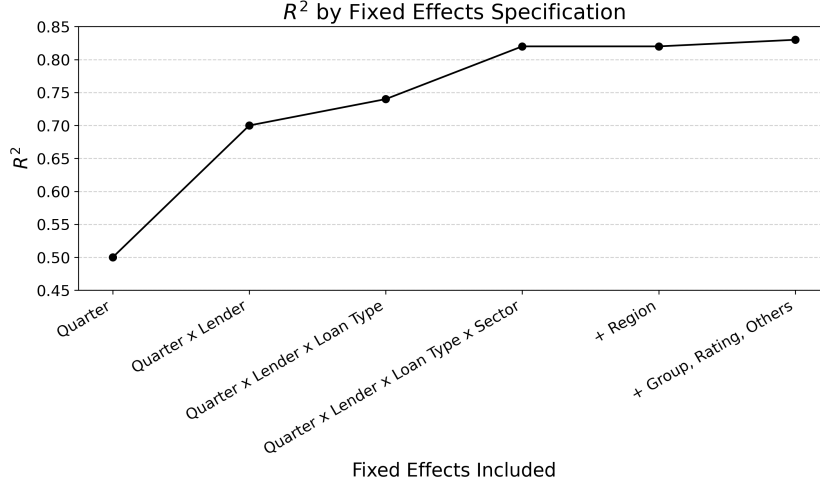


Figure 1: The figure shows the increase in R^2 of the baseline regression after gradually including fixed-effects interactions and the other observables. Observables explain up to 83% of the total variation in rates.

Fig. 1 summarizes the share of variation explained as I progressively add interacted fixed effects and other observables. Overall, observables explain 83% of the total variation in interest rates. Quarter-bank fixed effects alone explain 70%, while adding loan type times sector accounts for an additional 10 percentage points. The remaining observables contribute little additional explanatory power. The full regression table is reported in Table B.1 in the appendix.

I run alternative specifications in Appendix B: I experiment with other regression models, to account for the fact that by construction there are no rates below zero (Tobit, truncated regression), but with worse explanatory performance (Appendix B.2). In addition, I include a specification with firm fixed effects (Table B.2 and Fig. B.4) and one run separately by bank (Fig. B.5).

Time and cross-sectional distribution of rate residuals. I compute the standard deviation of residuals over time and across firm characteristics. Rate dispersion is substantial, 1pp on average over the time period — a meaningful gap in the context of loan pricing and external financing in general - and varies over the time period, as shown in Fig. B.2. Using monetary policy shocks, I also show that dispersion decreases after a rate hike.

Rate residuals and competition. I study the distribution of unexplained rate residuals in relation to standard measures of competition. Using credit registry data, I aggregate lending information across geographical areas and time. Specifically, I sum the amount of drawn credit for each firm at the bank-department-quarter level to assess the relative

exposure of each bank in that market, and I compute a standard Herfindahl-Hirschman Index (HHI) of concentration. Then I compute the standard deviation of residuals by quarter-department. I find that in more concentrated markets firms pay higher rates, but the distribution of rates is less dispersed (Fig. B.8).

3.3 From public credit ratings to default probabilities

Throughout the sample, I observe firms' annual rating reviews, with firms staying in the same class or transitioning to a different one. When a firm is upgraded to a safer rating class, it moves to a class in which firms have a lower expected probability of default. The objective now is to turn a rating update into a change in default probability.

Approach. I compute historical 3-year ahead default frequencies by rating class and year: I take all firms in the same rating category at 01/01/x and compute the share that defaulted by 31/12/x+2. The resulting default frequencies are reported in Fig. 2, which shows two important facts. First, ratings do very well at discriminating across firms' default risk: safer classes have consistently lower default frequencies; they are a reliable measure of credit risk. Second, default frequencies feature time variation, but the relative distance in default frequencies between classes is stable over time. This stability allows me to focus on changes in default probabilities arising from rating transitions, rather than also incorporating changes due with time-varying macroeconomic conditions, by fixing the default probability associated with each rating class to a specific year (I take 2021, the most recent before the rating reform). Specifically, for each firm i , I assign a default probability $P_{i,t}$ at time t as the default frequency associated to that class in 2021 and $\Delta P_{i,t+k} = (P_{i,t+k} - P_{i,t+k-1})$ is the annual default probability change, in percentage points, of firm i between $k - 1$ and k years after issuance.

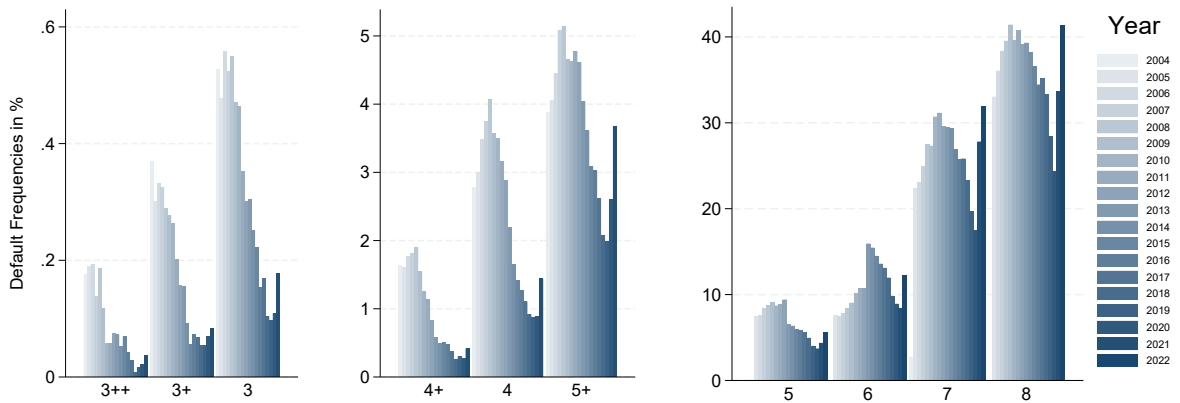


Figure 2: 3-year ahead default frequencies, by rating class and year (2004-2022).

Using rating transitions has three advantages over other measures of realized or perceived riskiness. First, it is an assessment of firms' default risk over a three-year horizon,

a more direct measure of creditworthiness than ex-post balance-sheet indicators or growth outcomes. Second, rating changes are far more frequent than defaults: firms are reviewed every year and can change class several times; for robustness, Appendix B.6.1 in the appendix repeats the test with defaults. Finally, this is a realized outcome rather than the reported subjective expectation of the bank. Other papers include banks' internal ratings to explain interest rates, but typically they do not test whether they effectively predict future default. This is an important validation exercise, as banks' internal assessments may be biased by regulatory considerations and capital requirements.

3.4 Rate residuals and changes in default probability

Consider the following regression:

$$r_{ilbt} = \alpha + \sum_{k=1}^K \beta_k \Delta P_{i,t+k} + \mathbf{x}'_{ilbt} \gamma + \mu_{bt\tau s} + \varepsilon_{ilbt}, \quad (2)$$

where r_{ilbt} is the rate on loan l granted by bank b to firm i in quarter t , $\mathbf{x}_{ilbt}$ and $\mu_{bt\tau s}$ are the same controls included when measuring rate residuals; $\Delta P_{i,t+k}$ is the annual default probability change, in percentage points, of firm i between $k-1$ and k years after issuance — this term is zero if the firm does not change rating class. I set the horizon $K = 10$ and use a balanced sample, so the same firms identify every β_k , and I exclude loans to firms that are already in default when the loan is granted ($P_{i,t} \neq 1$).

The main focus is to assess whether the rates contain forward information on changes in default probability, as summarized by $\beta_k \neq 0$. Two features of Eq. (2) matter for the reading of β_k . All K surprises enter jointly, so β_k is the correlation with the revision at horizon k holding the revisions at every other horizon fixed. And since the controls and the fixed effects sit in the same regression, β_k equals the coefficient of a regression of the rate residual on the rating surprise residual: it is a statement about the part of the rate the observables do not explain.

I plot the β_k in Fig. 3: the coefficients are positive and significant, stronger in the first year after the loan issuance and fading at longer horizons. This suggests that rates contain forward-looking information about the borrower's creditworthiness that is not captured by observable characteristics at the time of loan issuance. The fact that the correlation is stronger in the first year may indicate that banks have better information about short-term risks or that the impact of unobserved factors diminishes over time. I repeat the exercise using actual defaults in Appendix B.6.1. I reverse the regression by putting the rating revision on the left and the rate on the right, focusing on predicting the annual change, with and without a balanced panel, Appendices B.6.2 and B.6.3, or the cumulative change from loan issuance, Appendix B.6.4.

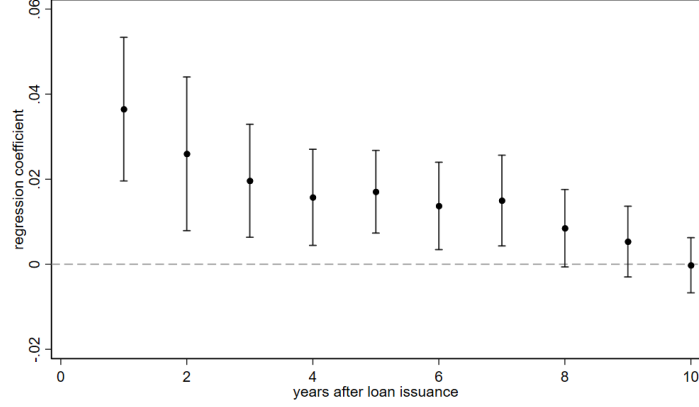


Figure 3: Correlation between rate residuals and unexpected changes in default probability, by year. Balanced sample.

3.4.1 Correlation across firms' demographics

I further explore how the forward information contained in rates varies across borrowers' demographics.

I separate firms by their Banque de France **rating** at loan issuance into three broad categories, and I document that the positive correlation is driven by firms with a relatively good rating, while it is not significant for relatively worse firms (see Fig. B.18). This finding might be counterintuitive, as riskier firms are closer to the default threshold.

One possibility is that interest rates may induce a worsening of the credit rating by mechanically increasing firm leverage, and in this case I should find a stronger effect when the loan is relatively large. By adding to Eq. (2) interaction terms between changes in default probabilities and **loan share of assets**, I find that the correlation coefficients when comparing the 10th and 90th percentiles of loans over assets are not significantly different (Fig. B.19). I repeat the exercise with the share of bank debt, in case the residual was informative about the rates on other loans, and results are robust. I move to **age interactions**, and I find that the correlation for younger firms (below the 10th percentile of age in my sample) is significantly larger than for older ones (above 90th percentile), as shown in Fig. B.20. Age is both a proxy for risk/growth prospects (Haltiwanger et al., 2013) and a proxy for time in the credit market (Cloyne et al., 2023), so this finding suggests that the correlation may be stronger when information is more valuable.

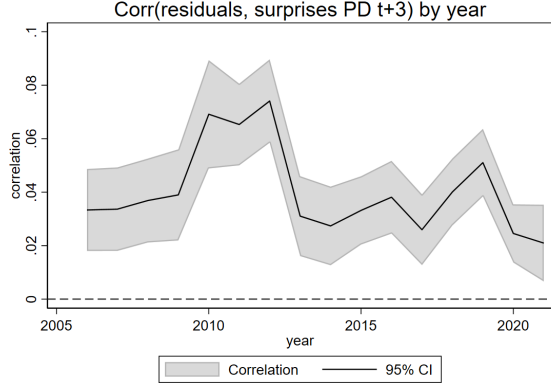
The above methods require many observations, so I rely on a different approach to examine the time dimension and other cross-sectional dimensions: I compute the rate

residuals $\tilde{r}_{ilbt}$ and the surprises in rating changes $\widetilde{\Delta P}_{ilbt}$ for the entire population:

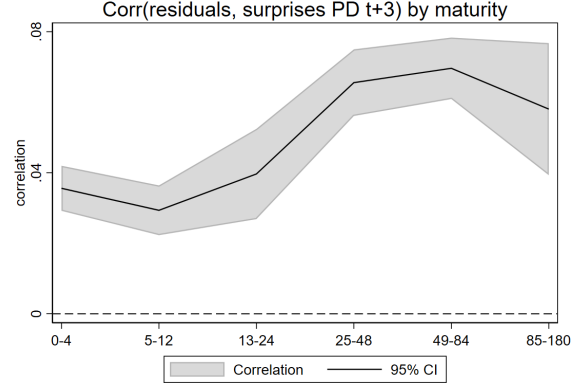
$$\begin{aligned} r_{ilbt} &= \alpha^r + \mathbf{x}'_{ilbt} \gamma^r + \mu_{bt\tau s}^r + \tilde{r}_{ilbt}, \\ \Delta P_{i,t \rightarrow t+3} &= \alpha^P + \mathbf{x}'_{ilbt} \gamma^P + \mu_{bt\tau s}^P + \widetilde{\Delta P}_{ilbt}. \end{aligned}$$

Then, I extract the pure correlation within each group of firms $\text{Corr}(\tilde{r}_{ilbt}, \widetilde{\Delta P}_{ilbt})$.

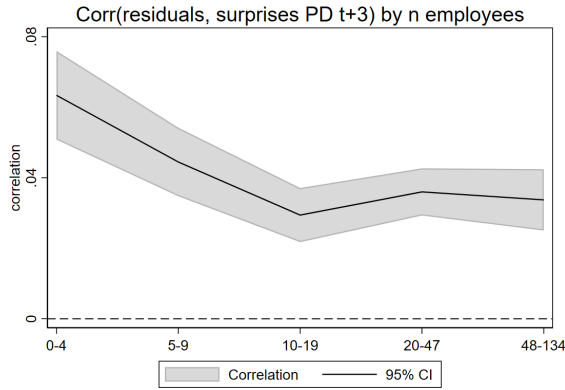
Results are shown in Fig. 4. The correlation varies over **time** (Fig. 4a), peaking around 2009–2012 at the height of the global financial crisis and sovereign debt crisis, a period of heightened uncertainty, and declining during the zero lower bound period. It rises with **loan maturity** up to the 25–84 month range, which also suggests that the value of information tends to be stronger at longer-horizons (Fig. 4b). Finally, the correlation is decreasing in firm size, as measured by **number of employees** (Fig. 4c) and age (Fig. 4d): the correlation for firms aged 0–4 years (≈ 0.06) is roughly twice that for firms aged 10 years or more, and similarly higher for firms with fewer than 10 employees relative to larger ones. Younger and smaller firms have shorter credit histories and less public information available.



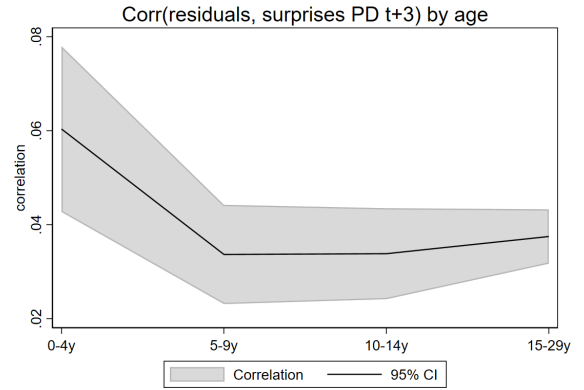
(a) By year



(b) By loan maturity (months)



(c) By number of employees



(d) By firm age

Figure 4: *Informational content of rate residuals across borrower and loan characteristics.* Each panel shows the correlation $\hat{\rho}_g$ between the unexplained rate component and the unexpected 3-year change in default probability (ΔPD_{t+3}), both computed in the entire population and correlated by group g . Shaded bands are 95% confidence intervals based on the Fisher z -transformation applied to $\hat{\rho}_g$.

Overall, these cross-sectional patterns suggest that the informational content of unexplained rate residuals is stronger when information is more valuable: for young and small firms, during the crisis and for longer contracts. In other words, the residuals may reflect *banks' information* about the evolution of firms' creditworthiness. The alternative explanation is that it is the result of a *leverage effect*, with higher rates inducing higher ex-post riskiness through the worsening of firms' fundamentals. Loan-size interactions in Fig. B.19 provide evidence against the leverage hypothesis. However, to rule it out, I adopt a quasi-experimental design, by comparing the effect of a rise in rates between firms borrowing at fixed and variable rates.

3.5 Test: do higher rates induce rating worsening?

I test the transmission of interest rate changes to firm-level risk. I exploit the effect of a raise of rates on default probability by comparing floating and fixed rate loans, to test whether firms borrowing variable rate loans are more likely to transition to a worse rating class when reference rates increase. I find no significant impact of increased debt-servicing costs on subsequent rating worsening, suggesting that an increase in interest rates inducing an increase in leverage cannot alone induce a worsening in firm-level risk in this sample. As the intensity of the effect depends on the individual firm’s exposure to variable rates, I further add an interaction term of the reference rate increase with the loan-to-assets and total bank debt-to-assets ratios, drawing from [Gürkaynak et al. \(2022\)](#) and [Core et al. \(2025\)](#); the result is unchanged. To further mitigate the issue of self-selection of firms into fixed or floating rate contracts, I repeat the exercise on loans originated at the end of the ZLB period, when firms did not expect a rate hike (end 2021), with same results.

I regress the change in default probability from loan origination t in year 1 to the end of year $t+3$ on cumulative increases in reference rates from t to $t+3$. I estimate several specification, including variable rate dummies, yearly increases in reference rates and their interactions with the share of the loan over assets, or bank debt over assets as proxies of the total share of variable rate loans of firms. The baseline specification is the following:

$$\Delta PD_{i,t+3} = \beta \Delta \text{Ref}_{l,t \rightarrow t+2} + \delta (\Delta \text{Ref}_{l,t \rightarrow t+2} \times \text{Share}_{i,t}) + \mathbf{x}'_{ilbt} \gamma + \mu_{bt\tau s} + \varepsilon_{ilbt}, \quad (3)$$

where $\Delta PD_{i,t+3} = 100 \times (P_{i,t+3} - P_{i,t})$ is the change, in percentage points, in the Banque de France default probability of firm i from loan issuance to the rating revision at year three; note that the rating review may happen at any month of year three. $\Delta \text{Ref}_{l,t \rightarrow t+2}$ is the overall change in loan l ’s own reference rate until the end of year 2: each floating loan is tied to a specific index (EONIA, Euribor 1m, 3m or 1y, TMO), and a fixed-rate loan is assigned a change of zero. $\text{Share}_{i,t}$ is either loan-over-assets or bank debt-over-assets, proxying the firm’s overall exposure to floating rates. $\mathbf{x}_{ilbt}$ holds what the bank observes at issuance: log loan size, loan over assets, maturity, dummies for a variable rate, a personal loan and an individual entrepreneur, firm age and its square, a dummy for a firm younger than three years, and, each scaled by total assets, short-term debt, undrawn credit, pledged guarantees, revenues, bank debt, non-bank debt and mobilised credit, plus region, turnover-class, rating-class and group-status fixed effects. $\mu_{bt\tau s}$ absorbs lender $\times$ month $\times$ loan-type $\times$ two-digit-sector effects, so β compares loans made by the same bank, in the same month, of the same type, to firms in the same industry. Identification therefore comes from variation in reference-rate exposure within such a cell: whether the

loan is fixed or floating and, among floating loans, which index it is tied to. Standard errors are two-way clustered by lender and by month. I report seven specifications: the floating-rate dummy alone, the cumulative rate increase alone, and the annual increases entered separately, each with and without the interaction in Eq. (3).

Appendix B.7.1 states the identifying assumptions, gives evidence on them, and reports the estimates, for the baseline sample and for loans issued in 2021q3–q4. Higher debt-servicing costs do not, on their own, make firms riskier: the correlation between rate residuals and future default documented above reflects what banks know at issuance, not what the rate itself does to the borrower.

4 Model

In the previous section I showed that interest rate dispersion, defined within banks and across firms, is informative of the degree of competition and screening in the banking sector. To quantify their relative role in determining observed dispersion, I introduce a model of loan pricing in which imperfect competition and asymmetric information are jointly at play. This model will be able to reproduce observed dispersion via imperfect competition (*within-type dispersion*) and screening (*between-type dispersion*), and the positive correlation between rates and risk when banks recognise riskier types and charge higher rates. In the next section, I will calibrate this framework to provide an estimate of the relative importance of screening and search in my framework.

The framework draws from the general market setting proposed by [Lester et al. \(2019\)](#), where competition is modeled via search frictions, and I adapt it to the bank loan market. However, I deviate from the assumption of menu contracts, for which I find little evidence in my sample³. Instead, I assume that banks can choose to acquire costly signals on firms true credit risk and adjust the interest rates, fixing quantity. For generality, I also introduce an interest rate “leverage effect”.

4.1 Environment

I examine the interactions between two homogeneous and risk-neutral banks and a continuum of firms. Firm sample banks’ offers to take one loan of fixed quantity 1 at an interest rate r to undertake a project. Projects are risky, characterized by a success probability p , which for generalization could be endogenous to the interest rate on the loan r - a higher interest rate linearly decreases the ex-post success probability:

$$p = p(r), \quad \text{with} \quad \frac{\partial p}{\partial r} < 0. \quad (4)$$

³I find that, conditional on all observable characteristics in my sample, there is a negative relationship between rates and quantity

Consequently, the bank’s effective cost of lending, denoted by c , becomes a function of the interest rate charged. As the rate rises, the default risk increases, implying:

$$c_i = c_i(r), \quad \text{with} \quad \frac{\partial c}{\partial r} > 0. \quad (5)$$

4.1.1 Gains from trade.

Banks. The opportunity cost of bank capital is set by the central bank’s policy rate, r^{CB} . I assume *limited liability*, so firms only repay the loan if the project is successful. Therefore, the competitive break-even rate for banks, c , is defined by the condition that the expected repayment equals the opportunity cost of capital $c = \frac{r^{CB}}{p(c)}$.

Firms. There is a continuum of risk-neutral firms of measure 1 per bank. I assume that firms have a reservation rate R , which for simplicity represents the value of the project if successful⁴, and that if they reject the rate offer they do not undertake the project and get a infinitely small, strictly positive, value $\varepsilon > 0$. In other words, firms will only accept a loan if the interest rate is below their reservation rate. I also assume that firms are affected by monetary policy rates exclusively through bank lending.

4.1.2 Market Frictions

The model departs from the standard competitive framework by introducing two distinct types of frictions: imperfect competition via search frictions and asymmetric information regarding firm types.

Search Frictions. I introduce search frictions as in the dispersed equilibrium framework of [Burdett and Judd \(1983\)](#) (BJ), which generates imperfect competition between banks and rate dispersion not related to credit risk. Specifically, I assume that a fraction π of firms are “shoppers”, meaning they sample offers from both banks and accept the lowest one, while the remaining fraction $1 - \pi$ are “captive”, meaning they only sample one offer and accept it if it is below their reservation rate. Banks know the aggregate proportion π , but the specific number of offers a firm has sampled is private information to the firm. For expositional convenience, I assume that π is constant across firm types, although this assumption can be relaxed. This structure leads to a mixed-strategy equilibrium in interest rates.⁵

Asymmetric Information. The population of firms is divided into two types, *Good* (share μ_g) and *Bad* (share μ_b), where $\mu_b + \mu_g = 1$, differing in riskiness, as the probability

⁴it may also represent the minimum rate available in the non-bank market

⁵The same dispersed equilibrium can be obtained with informed and uninformed buyers as in [Varian \(1980\)](#), or under sequential search as in [Stahl \(1989\)](#); see [Baye et al. \(1992\)](#) for the full set of mixed-strategy equilibria in this class of models. I adopt the sampling structure of [Burdett and Judd \(1983\)](#) because the share of shoppers π maps directly into a measure of competition that I can calibrate.

of success is lower for bad firms: $p_b(r) < p_g(r)$. Consequently, for banks the cost of lending to a bad firm is strictly higher than lending to a good firm at a given interest rate r : $c_g(r) < c_b(r)$. In the following, I also assume that the reservation rate of good firms is lower than that of bad firms: $\bar{r}^g < \bar{r}^b$. This assumption is not crucial to solve the model, but it will be important to deliver the positive correlation between the firm's type and the interest rate paid. The bank cannot directly observe the type of a firm, but it knows the population shares and the type-specific cost functions and reservation rates. This structure leads to lending to bad-types only or partially pooling equilibrium in interest rates, depending on the severity of the adverse selection problem.⁶ Finally, I look at the case in which $c_g(r) < c_b(r) < \bar{r}^g < \bar{r}^b$.

4.1.3 The information environment

Let $\theta \in \{g, b\}$ denote the firm's true type, with population shares μ_g and $\mu_b = 1 - \mu_g$. The market generates a binary signal $s \in \{g, b\}$ on each firm, which banks can consult at cost C_s per firm. Two features of this signal matter for what follows.

First, the signal is *public*: both banks observe the same realization s for a given firm, so they never disagree on how a firm is classified. What is hidden is the firm's true type θ , not the evidence about it; banks hold no informational advantage over one another, and the asymmetry runs between banks and firms rather than between banks. The signal is public among banks but not observed by the econometrician, which is why it can account for rate variation that observable characteristics leave unexplained.

Second, the precision of the signal is a *feature of the market*, not a decision of the banks. It measures how much can be learned about a borrower in a given market: the depth of credit registries and rating coverage, the length of firms' reporting histories, the disclosure that regulation imposes, the accounting and legal infrastructure that makes such evidence verifiable. Banks take this precision as given. Their only decision is whether to pay C_s and price on the signal, or to lend uninformed; I return to that decision in Section 4.3.

The signal is informative but imperfect:

$$\Pr(s = \theta \mid \theta) = q_\theta > 0.5, \quad \forall \theta \in \{g, b\}$$

where q_g is the probability of correctly identifying a good type, and q_b is the probability

⁶The pooling region and the exit of good types are the counterparts, in my setting, of the two costs of hidden quality identified by [Rothschild and Stiglitz \(1976\)](#) and [Stiglitz and Weiss \(1981\)](#); here, however, separation is obtained through the market's public signal rather than through a contractual screening device as in [Bester \(1985\)](#).

of correctly identifying a bad type.

The fraction of firms receiving each signal is:

$$\begin{aligned}\sigma_g &\equiv \Pr(s = g) = q_g \mu_g + (1 - q_b) \mu_b \\ \sigma_b &\equiv \Pr(s = b) = q_b \mu_b + (1 - q_g) \mu_g\end{aligned}$$

By construction, $\sigma_g + \sigma_b = 1$.

Bayesian updating. Upon observing the public signal $s_i \in \{g, b\}$, both banks update the common prior belief μ_g to the same posterior belief about facing a good type:

$$\begin{aligned}\alpha^{s_g} &\equiv \Pr(\theta = g \mid s = g) = \frac{q_g \mu_g}{q_g \mu_g + (1 - q_b) \mu_b} \\ \alpha^{s_b} &\equiv \Pr(\theta = g \mid s = b) = \frac{(1 - q_g) \mu_g}{(1 - q_g) \mu_g + q_b \mu_b}\end{aligned}$$

Note that $\alpha^{s_g} > \mu_g > \alpha^{s_b}$: a good signal increases the belief that the firm is good, while a bad signal decreases it. The complementary probabilities of facing a bad type are:

$$1 - \alpha^{s_g} = \frac{(1 - q_b) \mu_b}{q_g \mu_g + (1 - q_b) \mu_b}, \quad 1 - \alpha^{s_b} = \frac{q_b \mu_b}{(1 - q_g) \mu_g + q_b \mu_b}$$

4.2 Equilibrium Analysis

Throughout, I restrict attention to the class of *symmetric* equilibria: conditional on the signal, both banks post the same rate distribution $F(\cdot; q)$. The common precision $q \in (0.5, 1)$ is not itself a restriction — the signal is a public feature of the market, so both banks read the same evidence on every firm by construction — and the restriction bites on rate setting alone. All the equilibria characterized below are symmetric in this sense.

I show here the equilibrium properties of the model in the extreme cases of a perfect signal ($q = 1$) and of no information ($q = 0.5$); the general case, for any $q \in (0.5, 1)$, is solved in Appendix C.1. The equilibrium features *within-type* rate dispersion for any $\pi \in (0, 1)$, and may also feature *between-type* dispersion, depending on the information structure and on the degree of adverse selection.

4.2.1 Equilibrium under perfect signal ($q=1$)

The bank perfectly recognizes the type when it meets a firm. Basically, this is equivalent to solving independently two type-specific maximization problems.

Profit maximization. Banks' expected profits are as follows:

$$Profits^i(r) = \begin{cases} \mu_i[(1 - \pi) + \pi(1 - F(r))](r - c_i) & \text{if } r \leq \bar{r} \\ 0 & \text{otherwise} \end{cases}$$

That is, the bank trades-off higher profits and lower acceptance probability: every firm i is met with probability μ_i ; the bank gets $(r - c_i)$ if the firm accepts; the firm accepts with probability 1 if it is captive, and if it observes a higher offer from the other bank when it is a shopper, with probability $(1 - F(r))$. $F(r)$ is the equilibrium rate distribution.

In limit cases of monopoly ($\pi = 0$) or perfect competition ($\pi = 1$), the distribution is degenerate. The bank offers to each type the reservation rate $\bar{r}^i$ under monopoly, as all firms will always sample only one offer, and the competitive rate c_i under perfect competition, as all firms will visit both banks and get two rate quotes. However, under imperfect competition $\pi \in (0, 1)$, a dispersed rate equilibrium arises, as in BJ.

Exactly like in BJ, the type specific offered distribution is derived from the equal profit condition:

$$\mu_i(1 - \pi)(\bar{r}^i - c_i) = \mu_i[1 - \pi + \pi(1 - F^i(r))](r - c_i)$$

additionally assuming that $c'(\bar{r}^i) < 1$.

Proposition 4.1. *For $\pi \in (0, 1)$ and under perfect information (or perfect signal), there exists a unique symmetric mixed-strategy equilibrium⁷. For each firm type i , banks randomize their rate offers according to the cumulative distribution function $F^i(r)$ on the support $[\underline{r}^i, \bar{r}^i]$:*

$$F^i(r) = \begin{cases} 0 & \text{if } r < \underline{r}^i \\ 1 - \frac{1 - \pi}{\pi} \left[\frac{\bar{r}^i - r}{r - c_i(r)} \right] & \text{if } \underline{r}^i \leq r \leq \bar{r}^i \\ 1 & \text{if } r > \bar{r}^i \end{cases}$$

where the upper bound $\bar{r}^i$ is the reservation rate, and the lower bound $\underline{r}^i$ is implicitly defined by $F(\underline{r}_i) = 0$.

⁷Dispersion arises for the same reason as in [Burdett and Judd \(1983\)](#). With two perfectly symmetric banks, no rate strictly between the cost c_i and the reservation rate $\bar{r}^i$ can be sustained as a pure-strategy equilibrium: a bank posting it is undercut by ε by its competitor, who thereby captures the entire population of shoppers. Undercutting continues until the rate reaches the competitive level, where profits are zero; but there any bank can raise its rate and earn strictly positive profits on its captive firms, who accept any offer below $\bar{r}^i$. No pure strategy survives, so banks randomize, and the equilibrium distribution is pinned down by the requirement that expected profits be equal across the support. The same argument underlies each of the mixed-strategy equilibria characterized below.

and the observed rate distribution across types is the mixture of the distributions, weighted by each type's population share $\mu_g F^g(r) + \mu_b F^b(r)$ on the support $\underline{r}^g < r \leq \bar{r}^b$.

Dispersion and predictive power of residuals. Interestingly, the total dispersion of rates, measured as the total variance of the observed distribution, can be decomposed into a within-type, driven by the search friction only, and a between-type component, driven by the separation of the two types:

$$Var(R) = \mathbb{E}[Var(r | i)] + Var(\mathbb{E}[R | i]) = \sum_{i \in \{g, b\}} \mu_i Var(r | i) + \mu_g \mu_b (\mathbb{E}_g[r] - \mathbb{E}_b[r])^2$$

Secondly, the correlation between rates and default is positive, as the expected rate of bad types is higher than that of good types:

$$\rho(r, p) = \sqrt{\mu_g \mu_b} \cdot \frac{E[r|g] - E[r|b]}{\sqrt{Var(r)}}$$

4.2.2 Equilibrium under uninformative signal ($q=0.5$)

The bank operates under information asymmetry, namely it cannot distinguish the firm type upon meeting. I may still assume that the bank knows the population composition and the type-specific costs and reservation rates. Consequently, it faces a type-specific demand on two separate intervals: if $r < \bar{r}^g$, both types would accept an offer (*pooling region*), while if $\bar{r}^g < r < \bar{r}^b$ the bank knows that only bad types would accept (*bad-only region*) — a dispersed version of the lemons outcome of [Akerlof \(1970\)](#). The equilibrium may feature a mix of lending on the pooling and bad-only region depending on the severity of the adverse selection problem; mild adverse selection would incentivize pooling, while in the severe adverse selection case banks are more likely to reduce or avoid pooling and offer only high rates.

Limit cases of monopoly/competition $\pi = 0, 1$. I begin by analyzing the equilibrium in the extreme cases of monopoly and perfect competition to build the intuition for the general case.

Monopoly $\pi = 0$. The bank makes profits

$$\Pi = \begin{cases} (r - c^{avg}(r)) & \text{if } \underline{r} < r \leq \bar{r}^g \\ \mu_b(r - c_b(r)) & \text{if } \bar{r}^g < r \leq \bar{r}^b \end{cases}$$

where $c^{avg}(r) = \mu_g c_g(r) + \mu_b c_b(r)$ is the weighted average of the type-specific costs. The bank compares the maximum profit it can extract in each region:

$$\max\{(\bar{r}^g - c^{avg}(\bar{r}^g)), \mu_b(\bar{r}^b - c_b(\bar{r}^b))\}$$

I assume that marginal costs increase sufficiently slowly ($c'(r) < 1$) such that profits are increasing in rates within each interval⁸. Consequently, the optimal strategy must be a corner solution: the bank compares the maximum profit from pooling (at $\bar{r}^g$) against the maximum profit from screening (at $\bar{r}^b$).

Lemma 4.1. Define ϕ :

$$\phi \equiv (\bar{r}^g - c^{avg}(\bar{r}^g)) - \mu_b(\bar{r}^b - c_b(\bar{r}^b))$$

Under monopoly ($\pi = 0$), $c'(r) < 1$ and with an uninformative signal, the unique equilibrium is $\bar{r}^g$ if $\phi > 0$, and $\bar{r}^b$ if $\phi \leq 0$.

Under monopoly, the unique profit-maximizing interest rate depends on the sign of ϕ . **I can interpret ϕ and an indicator of the strength of asymmetric information.** If $\phi \leq 0$, adverse selection is severe, and banks extract more profits from trading with bad types only. If $\phi > 0$ adverse selection is mild, and banks extract more profits from pooling good and bad types.

Perfect competition $\pi = 1$ In this case, the bank makes zero profits, offering the competitive rate in the population.

Lemma 4.2. *Under perfect competition ($\pi = 1$) and asymmetric information, the unique equilibrium is a pooling contract at the competitive rate $r^c = c^{avg}$, implicitly defined by the break-even condition given average costs. Given the assumption $c^b < \bar{r}^g$, this rate is strictly below the reservation rate of both types and is accepted by the entire population.*

When $\pi = 1$, firms always compare offers. Bertrand competition drives the interest rate down to the break-even point. Since the competitive rate satisfies $r^c < \bar{r}^g$ by assumption, both types participate, and the bank faces the average cost of the population. The equilibrium profit is therefore $\Pi = (\mu_b + \mu_g)(r^c - c^{avg}(r^c)) = 0$.

General case: Imperfect Competition. Bank profits depend on the interval.

$$\Pi = \begin{cases} [1 - \pi + \pi(1 - F(r))](r - c^{avg}(r)) & \text{if } r \leq \bar{r}^g \\ \mu_b[1 - \pi + \pi(1 - F(r))](r - c_b(r)) & \text{if } \bar{r}^g < r \leq \bar{r}^b \end{cases}$$

⁸if $c'(r)$ was increasing over the rate interval, there may be a point where the cost is higher than the benefit, and an internal solution would be optimal ($r^* < \bar{r}^g$ or $r^* < \bar{r}^b$)

where $c^{avg}(r) = \mu_g c_g(r) + \mu_b c_b(r)$. As in the perfect information case, the bank trades-off profits and acceptance probability. However, the bank knows that both firms would accept a rate in the pooling region, while a rate on the bad-only region would be accepted by bad types and rejected by good type, that would exit the market. As in the perfect information case with one type, the equilibrium CDF will be derived as the isoprofit function given profits at the maximum rate charged. However, now the maximum rate depends on the severity of adverse selection, as captured by the value of ϕ .

Severe adverse selection: $\phi \leq 0$

Banks surely trade with bad types, and possibly offer some rates in the pooling region. When trading with captive firms, the bank makes more profits from trading with bad types than pooling. Therefore, the maximum rate observed in the market is $\bar{r}^b$. The bank offers a distribution of rates on the bad-only region to extract profits from bad types, and may also offer some rates in the pooling region to extract profits from good types, such that:

$$\begin{aligned} [1 - \pi + \pi(1 - F^a(r))](r - c^{avg}(r)) &= \mu_b(1 - \pi)(\bar{r}^b - c_b(\bar{r}^b)) & \text{if } r \leq \bar{r}^g \\ \mu_b[1 - \pi + \pi(1 - F^a(\bar{r}^g) - F^b(r))](r - c_b(r)) &= \mu_b(1 - \pi)(\bar{r}^b - c_b(\bar{r}^b)) & \text{if } \bar{r}^g < r \leq \bar{r}^b \end{aligned}$$

Proposition 4.2. *For $\pi \in (0, 1)$, $\phi \leq 0$, $c'(r) < 1$ and with an uninformative signal, there exists a unique symmetric mixed-strategy equilibrium. Banks randomize their rate offers according to the cumulative distribution function $F(r)$ on the support $[\underline{r}^a, \bar{r}^b]$:*

$$F(r) = \begin{cases} 0 & \text{if } r < \underline{r}^a \\ F^a(r) & \text{if } \underline{r}^a \leq r \leq \bar{r}^g \\ F^a(\bar{r}^g) + F^b(r) & \text{if } \bar{r}^g < r \leq \bar{r}^b \\ 1 & \text{if } r > \bar{r}^b \end{cases}$$

where the equal profit condition on each interval gives $F^a(r), F^b(r)$:

$$\begin{aligned} F^a(r) &= \max \left\{ 1 - \frac{1 - \pi}{\pi} \left[\frac{\mu_b(\bar{r}^b - c_b(\bar{r}^b))}{(r - c^{avg}(r))} - 1 \right], 0 \right\} \\ F^b(r) &= 1 - F^a(\bar{r}^g) - \frac{1 - \pi}{\pi} \left[\frac{(\bar{r}^b - c_b(\bar{r}^b))}{(r - c_b(r))} - 1 \right] \end{aligned}$$

and the lower bounds $\underline{r}^a$ and $\underline{r}^b$ are implicitly defined by $F^a(\underline{r}^a) = 0$ and $F^b(\underline{r}^b) = 0$.

$$\begin{aligned}\underline{r}^a &= c^{avg}(\underline{r}^a) + \mu_b(1 - \pi)(\bar{r}^b - c_b(\bar{r}^b)) \\ \underline{r}^b &= c_b(\underline{r}^b) + \frac{\bar{r}^g - c^{avg}(\bar{r}^g)}{\mu_b}\end{aligned}$$

This form ensures that the combined function on the two intervals cumulates to 1: by construction, $F^a(\bar{r}^g) + F^b(\bar{r}^b) = 1$. My equilibrium differs from [Burdett and Judd \(1983\)](#) in accounting for demand heterogeneity, preserving the mixed-strategy equilibrium result. On the existence of a pooling region, I find that it depends on the severity of adverse selection, as captured by ϕ .

Lemma 4.3. *The pooling region exists when adverse selection is relatively mild $-\frac{\pi}{1-\pi}(\bar{r}^g - c^{avg}) < \phi \leq 0$, while for values of ϕ below this threshold the bank lends to bad types only.*

I find the threshold from $F^a(\bar{r}^g) > 0$. Intuitively, when adverse selection is severe, the bank prefers to offer only bad-only contracts, as the profits from pooling are too low. When adverse selection is mild, the bank offers some pooling contracts to extract profits from good types, while still offering some bad-only contracts to extract profits from bad types.

Weak Adverse Selection: $\phi > 0$

Banks trade with both types (Pooling Equilibrium). Banks make more profits from pooling both firms at the reservation rate of good types when meeting captive firms. Consequently, the maximum rate observed in the market is $\bar{r}^g$, trades will only happen in the pooling region and all firms will receive credit. The CDF of rates is found by setting the equal profit condition on the pooling region:

$$[1 - \pi + \pi(1 - F^a(r))](r - c^{avg}(r)) = (1 - \pi)(\bar{r}^g - c_{avg}(\bar{r}^g))$$

Proposition 4.3. *For $\pi \in (0, 1)$, $\phi > 0$, $c'(r) < 1$ and with an uninformative signal, there exists a unique symmetric mixed-strategy equilibrium. Banks randomize their rate offers according to the cumulative distribution function $F(r)$ on the support $[\underline{r}, \bar{r}^g]$:*

$$F(r) = \begin{cases} 0 & \text{if } r < \underline{r} \\ 1 - \frac{1 - \pi}{\pi} \left[\frac{(\bar{r}^g - r)}{(r - c^{avg}(r))} \right] & \text{if } \underline{r} \leq r \leq \bar{r}^g \\ 1 & \text{if } r > \bar{r}^g \end{cases}$$

Where the upper bound $\bar{r}^g$ is the reservation rate of the good type, and the lower bound $\underline{r}$ is implicitly defined by $F(\underline{r}) = 0$.

With mild adverse selection, the equilibrium is closest to the BJ dispersed price equilibrium, as banks offer only pooling contracts.

4.2.3 Equilibrium under informative signal

I solve the general case of the equilibrium under any precision q . The general case is summarized in Appendix C.1. Banks face two subpopulations of firms receiving a i=good/bad signal. They offer a signal-specific distribution, knowing that the probability of good/bad types is imprecise. As the signal becomes more precise, banks' error is smaller and the rate distribution converges to the perfect information one. Denote the expected profit under any precision q as $\mathbb{E}[\Pi(q)] = \sigma_g \cdot \Pi^{sg} + \sigma_b \cdot \Pi^{sb}$, where Π^{sg} and Π^{sb} are the expected profits conditional on receiving a good or bad signal, respectively. The expected profit under no information is denoted as $\mathbb{E}[\Pi(q = 0.5)]$.

4.3 Implications for the value of information and surplus sharing

Total surplus (per bank). Under perfect information or perfect signal, per-trade surplus between a bank and a firm of type i is $(\bar{r}^i - c_i)$. In the symmetric equilibrium, only half of the π shoppers will effectively accept the bank's offer, while the others will prefer the second offer. The potential surplus attainable per bank if all firms are matched is $TS^{q=1} = (1 - \frac{\pi}{2})(\mu_b(\bar{r}^b - c_b) + \mu_g(\bar{r}^g - c_g))$. However, under asymmetric information total surplus depends on the severity of adverse selection: if adverse selection is severe, at least a fraction of good firms obtains a rate quote beyond their reservation rate, thus refuses and exits the market; the gains from trade remain unrealized.

Proposition 4.4 (Banks' share of total surplus). *Banks share of total surplus is decreasing in competition and increasing in the precision of the signal.*

Regime at $q = 0.5$	Π gain at $q = 1$	The gain is the surplus ...
Full pooling, $\phi > 0$	$(1 - \pi) \mu_b (\bar{r}^b - \bar{r}^g)$	... not extracted from <i>bad</i> types
Hybrid, $\underline{\phi} < \phi \leq 0$	$(1 - \pi) \mu_g (\bar{r}^g - c_g)$	... mixed, equal to the <i>good</i> types that exit
Bad-only, $\phi \leq \underline{\phi}$	$(1 - \pi) \mu_g (\bar{r}^g - c_g)$	... of the <i>good</i> types that exit

Table 2: Bank's gain from moving to a perfect signal, $\Delta\Pi \equiv \Pi^{q=1} - \Pi^{q=0.5}$, per firm approached by one bank.

Bank's share of total surplus. Under perfect information, the bank gets a share of realised surplus per trade exactly equal to $1 - \pi$, while firms get the remaining π share. Under asymmetric information, banks' profits are lower: in pooling, the bank

loses $(1 - \pi)(\bar{r}^b - \bar{r}^g)$ from each bad type with respect to the perfect information case; under bad-only lending, the banks do not get the share of surplus from good firms that exit the market.

Proposition 4.5 (Firms' surplus and information). *Information reduces (increases) costs borne by good (bad) types.*

- (i) *Under no information, when adverse selection is mild ($\phi > 0$), good types trade at the higher pooling rates; when adverse selection is severe, ($\phi \leq 0$), they may not trade at all. These gains are increasing in information q and decreasing in the level of competition.*
- (ii) *On the contrary, bad types benefit from pooling when adverse selection is mild ($\phi > 0$), as they trade at rates below the good type reservation rate; when adverse selection is severe $\underline{\phi} < \phi \leq 0$, they partially gain from cross-subsidization, but less than in pooling; they do not gain if adverse selection is very severe and banks. These benefits are decreasing in information q .*

Firms' share of total surplus. Good types bear two costs of asymmetric information: a cross-subsidy to bad types where the bank pools, and exclusion where the bank lends to bad types only. Both fall with signal precision. Under no information and $\phi > 0$ the bank pools good and bad types. Total cross-subsidy to bad types is $\pi\mu_b(c_b - c_g)$. Under no information and $\phi > 0$, $1 - F(\bar{r}^g)$ are offered a too high rate, and exit the market. if $F(\bar{r}^g) > 0$, the remaining good types, if any, subsidize bad types.

Regime at $q = 0.5$	ΔS_g	ΔS_b
Full pooling, $\phi > 0$	$\mu_g \frac{\pi}{2} \mu_b (c_b - c_g) > 0$	$-\mu_b \left[(1 - \pi)(\bar{r}^b - \bar{r}^g) + \frac{\pi}{2} \mu_g (c_b - c_g) \right] < 0$
Hybrid, $\underline{\phi} < \phi \leq 0$	no closed form, > 0	no closed form, ≤ 0
Bad-only, $\phi \leq \underline{\phi}$	$\mu_g \frac{\pi}{2} (\bar{r}^g - c_g) > 0$	0

Table 3: Firms' gain from moving to a perfect signal, $\Delta S_i \equiv S_i^{q=1} - S_i^{q=0.5}$, per firm approached by one bank; notation as in Table 2. Good types gain in every regime, bad types never gain. Under full pooling information only redistributes, $\Delta \Pi + \Delta S_g + \Delta S_b = 0$: the good types' gain is exactly the cross-subsidy $\mu_g \frac{\pi}{2} \mu_b (c_b - c_g)$ they stop paying, and the bank's gain is exactly what the bad types stop receiving. Under bad-only lending bad types are unaffected, since they already face their own cost c_b and their own reservation rate $\bar{r}^b$; the whole gain is good types re-entering the market. Total surplus rises by $\mu_g(1 - \lambda) \left[(1 - \pi) + \frac{\pi}{2}(1 - \lambda) \right] (\bar{r}^g - c_g)$, which is zero under full pooling and maximal under bad-only lending. The hybrid entries have no closed form because the equal-profit condition pins the bank's total profit but not its split across types; their signs are bracketed by the two adjacent rows.

Proposition 4.6 (Paying for the market signal). *In a market whose public signal has precision q , banks pay C_s and price on the signal if and only if the expected profit gain with respect to lending uninformed covers the cost:*

$$\mathbb{E}[\Pi(q)] - \mathbb{E}[\Pi(q = 0.5)] \geq C_s$$

where $\mathbb{E}[\Pi(q)]$ is the expected profit when banks price on a signal of precision q , and $\mathbb{E}[\Pi(q = 0.5)]$ is the expected profit when they lend uninformed. Otherwise both banks lend uninformed.

Banks do not choose q : the precision on offer is set by the market. The only margin is participation, that is, whether the information the market makes available is worth its price.

Comparing a market whose signal is perfect ($q = 1$) with one where no information is available ($q = 0.5$):

$$\begin{aligned} \mu_b(1 - \pi) (\bar{r}^b - \bar{r}^g) &\geq C_s && \text{if } \phi > 0 \\ \mu_g(1 - \pi) (\bar{r}^g - c_g(\bar{r}^g)) &\geq C_s && \text{if } \phi \leq 0. \end{aligned}$$

In words, the bank pays for information only if the profit that asymmetric information leaves on the table is larger than its cost. The source of the forgone profit differs by regime: under mild adverse selection ($\phi > 0$), it is the surplus the bank fails to extract from *bad* types; under severe adverse selection ($\phi \leq 0$), it corresponds to the surplus from *good* types, who face a rate above their reservation value and exit the market. Importantly, in both cases **banks' willingness to pay for the market's information is inversely proportional to the degree of competition π** .

Corollary 4.1 (Competition and the value of information). *At all levels of q , $\mathbb{E}[\Pi(q)] \propto (1 - \pi)$, the share of captive firms. The gain from pricing on a more precise signal is therefore also proportional to the share of captive firms. The value of information falls one for one with competition and vanishes as $\pi \rightarrow 1$: under perfect competition no bank pays a positive cost $C_s > 0$ to consult the signal, however precise the market makes it. Information can be abundant and still go unused.*

Because the signal is public, both banks read the same evidence on every firm, so the common precision q requires no further equilibrium selection.

5 Calibration

The next step is to give a quantitative assessment of the relative contribution of search and information to explained variance, along with the surplus-sharing trade-offs. Because

the structural parameters, π for competition and q for the signal precision, are inherently latent, I employ a calibration exercise to recover these values by matching the model to specific moments of the residual distribution across Banque de France rating classes. For each rating class, I calibrate three parameters: the shopper share π , which indexes the intensity of competition, the signal precision q , and the reservation rate for good types $\bar{r}_g$. The central bank funding cost r_{CB} is fixed at 1, consistent with the average policy rate over the sample period. Four additional parameters are calibrated directly from the data: $\mu_b, p_g, p_b, \bar{r}_b$. I observe the evolution of firms across rating classes and label as good or bad those that transition to a better or worse class. Firms that default outright are not counted as bad under this definition; Appendix E reports the alternative that includes them, and shows that the estimates below are unchanged for ratings 3+ through 5+. From this classification I derive the population share of bad types μ_b and the average success probabilities p_g and p_b in each population (computed as one minus the three-year-ahead default frequencies from Banque de France historical data), and the upper bound $\bar{r}_b$ (set at the 99th percentile of the observed rate distribution in each rating class, a more outlier-robust analogue of the literal maximum). For each rating class, I recentre the residuals on the class mean rate, so that all moments are expressed in rate levels rather than in deviations from the class mean.

I view this exercise as illustrative rather than as a literal structural estimation of the full cross-sectional distribution of borrower types. The model of Section 5 characterizes equilibrium for a stylized environment with exactly two borrower types, good and bad; the data, by contrast, reflect a continuum of borrower heterogeneity within each rating class. I do not expect, nor aim for, the two-type model to replicate every empirical moment exactly: my goal is to show that the search-and-screening mechanism generates rate dispersion and a rate–default correlation of the right sign and the right qualitative pattern across the credit-quality spectrum, using the two-type model as a tractable benchmark for the richer, continuum-type environment that plausibly governs the data. Accordingly, I treat the correlation and mean moments as the primary targets: the correlation is matched essentially exactly in every rating class, and the mean to within roughly 15% (Table 5); the variance moment, which is mechanically the most sensitive to how many latent types the model allows for, is matched only partially, and should be read as a lower bound on the dispersion the search-and-screening channel alone can generate rather than a claim that these two channels are the only source of within-class rate dispersion in the data. I return to this point after presenting the results.

5.1 Calibration Strategy

The calibration proceeds in two stages. First, I conduct a grid search over the parameter space, evaluating 20^3 combinations of parameter values within economically plausible bounds: $\pi \in [0.05, 0.95]$, $q \in [0.5, 1.0]$, and $\bar{r}_g \in [0.5 \bar{r}_{\text{mean}}, 0.98 \bar{r}_b]$, where $\bar{r}_{\text{mean}}$ is the class

mean rate. For each combination, I compute the model-implied moments—mean rate, variance, and correlation with subsequent default probability changes. Because a fraction π of firms compare offers from both banks and accept the lower one, the equilibrium distribution $F(r)$ derived in Section 5 – each bank’s own posted-rate strategy – is not the same object as the distribution of *realized*, originated rates: a firm who shops around ends up paying the better of two draws from F , not a raw draw from it. I therefore compute all targeted moments under the properly reweighted realized-rate distribution $g(r) \propto f(r)(1 - \pi F(r))$, derived in Appendix D.1, rather than under $F(r)$ directly. I keep the combination that minimizes the objective function, which is the weighted sum of squared percentage errors of targeted moments. Second, I refine the best grid point using local optimization (L-BFGS-B algorithm) to minimize the weighted sum of squared percentage deviations between model and data moments.

The objective function to minimize is the sum of squared proportional errors across the three moments⁹:

$$Q(\theta) = \left(\frac{m_{\text{mean}}^{\text{model}} - m_{\text{mean}}^{\text{data}}}{m_{\text{mean}}^{\text{data}}} \right)^2 + \left(\frac{m_{\text{var}}^{\text{model}} - m_{\text{var}}^{\text{data}}}{m_{\text{var}}^{\text{data}}} \right)^2 + \left(\frac{m_{\text{corr}}^{\text{model}} - m_{\text{corr}}^{\text{data}}}{m_{\text{corr}}^{\text{data}}} \right)^2 \quad (6)$$

With three moments and three free parameters (r_{CB} fixed), the order condition for exact identification is satisfied. The parameters must also generate a feasible equilibrium (costs below reservation rates) and satisfy the mixed-strategy conditions, which impose additional structure beyond the three explicit moments matched. Because the equilibrium is piecewise in $(\pi, q, \bar{r}_g)$, however, a formal rank condition for global identification is not analytically tractable; Section 5.4 reports numerical evidence in its place.

5.2 Counterfactual Analysis

To decompose the relative contributions of search frictions and information acquisition, I compute counterfactual equilibria holding recovered parameters $(\hat{\pi}, \hat{\bar{r}}_g)$ and calibrated parameters $\mu_b, p_g, p_b, \bar{r}_b, r_{CB}$ fixed while varying the signal precision q . In particular, I compare the percentage increase in variance given the calibrated parameters against the counterfactual with $q = 0.5$ (no information) isolates the role of search frictions:

$$\text{Information Contribution} = \frac{\text{Var}(\hat{q}) - \text{Var}(q = 0.5)}{\text{Var}(q = 0.5)} \times 100\% \quad (7)$$

5.3 Calibration Results

Table 4 presents the calibrated parameters for each rating class; Table 5 reports the corresponding model fit. The correlation moment is matched closely across all classes,

⁹proportional errors are used to ensure that moments of different scales are weighted comparably in the estimation.

and mean is matched to a reasonable approximation. Variance is matched only partially in every rating class; consistent with the two-type-versus-continuum-heterogeneity gap discussed above, I do not read this as a failure of the search-and-screening mechanism, and return to it below and in Section 5.4.

Table 4: Search Frictions and Information by Rating Class

Rating	$\hat{\pi}$	$\hat{q}$	$\hat{r}_g$	Info Contr. (%)	Obj. Value $Q(\theta)$
3+	0.772	0.997	3.461	4.89	0.41
3	0.759	1.000	3.706	4.56	0.35
4+	0.738	0.999	4.609	3.94	0.15
4	0.757	1.000	5.056	5.06	0.19
5+	0.749	1.000	5.403	3.05	0.20
5	0.729	0.944	6.662	0.04	0.04
6	0.721	0.526	7.057	0.00	0.04
7	0.711	0.526	9.350	0.00	0.00

Notes: $\hat{\pi}$ is the search-friction parameter, $\hat{q}$ the signal precision, $\hat{r}_g$ the reservation rate of good borrowers, and “Information Contribution” measures the percentage increase in rate dispersion attributable to screening relative to the no-information benchmark.

Table 5: Model Fit: Observed vs. Model-Implied Moments

Rating	Mean		Variance		Correlation	
	Data	Model	Data	Model	Data	Model
3+	1.67	1.94	0.406	0.154	0.025	0.025
3	1.83	2.08	0.443	0.188	0.033	0.033
4+	2.40	2.53	0.553	0.338	0.032	0.032
4	2.37	2.63	0.730	0.424	0.048	0.048
5+	2.57	2.80	0.867	0.492	0.048	0.048
5	3.36	3.44	0.985	0.794	0.017	0.017
6	3.63	3.67	1.137	0.913	0.000	0.000
7	4.79	4.78	1.843	1.725	0.000	0.000

Notes: Mean is the class mean rate and variance the variance of rate residuals within the class, both in percentage points. Correlation is between the rate residual and the three-year-ahead change in the borrower’s default probability. For ratings 6 and 7 the estimated correlation is not significantly different from zero ($p = 0.65$ and $p = 0.54$) and is set to zero as a calibration target.

The calibration results reveal three key findings. First, search frictions are relatively low across all rating classes, with $\hat{\pi}$ estimated between $[0.71, 0.77]$, increasing over ex-ante safety. This suggests that the French corporate lending market is relatively competitive, with most firms sampling multiple lenders—consistent with the prevalence of multi-bank relationships in the sample: on average about 60% of firms across classes maintain at least two active bank relationships (Table D.7).

Second, screening intensity exhibits heterogeneity across the credit quality distribution. Good borrowers (ratings 3+ through 5+) face near-perfect screening ($\hat{q}$ close to 1),

while poor borrowers (ratings 6 and 7) face essentially no screening ($\hat{q} \approx 0.53$).

Third, despite high screening intensity implied by the data moments for good ratings, information contributes minimally to observed rate dispersion. The information contribution to variance never exceeds 5.1%, indicating that search frictions account for most of unexplained dispersion even in segments with near-perfect screening. This partly reflects the mild severity of the adverse selection problem within each rating class, for example, the relative low distance of default probabilities between types mechanically limit between-type variance.

The objective values in Table 4 are highest for the safest classes and fall with credit quality, from 0.41 at rating 3+ to essentially zero at rating 7. The pattern is mechanical: in the safest classes the two types differ hardly at all in cost — $c_b - c_g$ is about 0.003 at rating 3+, against an observed standard deviation of rate residuals of 0.64 — so between-type dispersion is negligible and the model has to generate nearly all observed variance from search alone. The estimated $\hat{r}_g$ sits just below the class upper bound $\bar{r}_b$ in every class, so the pooling region covers most of the support. Appendix E reports the same two tables under the alternative bad-type definition that counts outright defaults as bad.

5.4 Robustness of the Estimated Equilibrium

The response to each posterior signal realization depends endogenously on the estimated parameters through the threshold conditions governing ϕ_{good} and ϕ_{bad} — an analytical proof that the three-parameter GMM estimator $(\pi, q, \bar{r}_g)$ is globally identified from the three targeted moments (mean, variance, and the rate–default correlation) is not tractable. I therefore rely on three complementary numerical checks, run separately for each of the eight rating classes, to build confidence that the parameter vector reported by my estimation procedure is the unique — or at least an isolated, dominant — minimizer of the GMM objective. Appendix D documents each of these three checks, and their complete numerical results, in full.

First, I assess local identification via the standard rank condition (Rothenberg, 1971): at the estimated parameter vector $\hat{\theta}$ I compute the 3×3 Jacobian of the three matched moments with respect to $(\pi, q, \bar{r}_g)$ by numerical differentiation. In every rating class the Jacobian has full rank (3/3), ruling out exact local non-identification, as summarized in Table 6. The associated condition numbers are not uniform, however: they range from roughly 5 to over 400 across rating classes, with the weakest direction consistently loading on q (screening precision) — most pronounced for the safest classes, whose near-perfect estimated precision ($\hat{q}$ close to 1) leaves the moments comparatively insensitive to q . I report this pattern transparently rather than treating identification as uniformly strong: q is comparatively less sharply pinned down than π and $\bar{r}_g$ in some classes, even though the local rank condition is always satisfied.

Second, because local curvature alone cannot rule out a second, disconnected mini-

Table 6: Local Identification Diagnostics: Elasticity-Scaled Jacobian of (mean, var, corr) w.r.t. $(\pi, q, \bar{r}_g)$

Rating	Rank	Condition Number	Weakest Direction
3+	3/3	431.7	q
3	3/3	209.4	q
4+	3/3	134.2	q
4	3/3	82.1	q
5+	3/3	65.9	q (mixed with $\bar{r}_g$)
5	3/3	4.8	π (mixed with $\bar{r}_g$)
6	3/3	21.6	q
7	3/3	14.1	q

Notes: Rank and condition number computed from the singular value decomposition of the elasticity-scaled Jacobian (percentage change in moment per percentage change in parameter) at each rating class’s estimated $\hat{\theta}$. “Weakest direction” identifies the parameter with the largest loading on the smallest singular vector; a second parameter is named when its loading also exceeds 0.45 in absolute value.

num elsewhere in the parameter space, I complement it with a dense search over the full admissible cube of $(\pi, q, \bar{r}_g)$ for each rating class, visualized directly as three-dimensional plots of the parameter combinations whose implied moments come within a chosen tolerance of the data. Within every equilibrium regime realized on the grid, the set of parameter combinations that adequately fit the data forms a single connected neighborhood around one point rather than several disjoint clusters — visual evidence against multiple, unrelated local optima within a given regime.

Third, I exploit the fact that the equilibrium’s regime-switching structure (as derived in Section 5) is itself governed by two threshold conditions — one for the pool of borrowers assigned a favorable signal, one for the pool assigned an unfavorable signal — each admitting three possible states (full pooling, hybrid, or bad-only lending), for nine theoretically possible regime pairs. Rather than assume ex ante which pairs are empirically relevant, I determine this data-drivenly: for each rating class I tag every point of the parameter grid with its actually realized (good-pool, bad-pool) regime pair, and find that only a subset — between three and six, depending on the rating class — are ever realized within the admissible parameter region, always respecting a simple ordering in which the good-signal pool is never in a “less pooled” regime than the bad-signal pool. I re-solve the GMM objective separately within each empirically realized regime and compare the best fit achieved across all of them. In every rating class, the best-fitting parameter vector across all admissible regimes coincides with the parameter vector reported by my baseline, regime-unconstrained estimation. This confirms that the baseline optimizer converges to the true global optimum of the objective function and is not trapped in a locally optimal equilibrium regime.

Taken together, these three checks — local curvature (rank and condition number of

the moment Jacobian), a dense global search of the parameter space visualized directly, and an exhaustive comparison across every equilibrium regime the model can produce — provide converging numerical evidence that the reported parameter estimates for each rating class correspond to a well-isolated global optimum of the GMM criterion. They do not constitute a formal identification proof, which the model’s regime-switching structure precludes in closed form, but they substantially reduce the concern that my point estimates are an artifact of the numerical optimization routine rather than a genuine feature of the data.

6 Surplus sharing and cross-subsidies

I return to the calibrated model of Section 5 to ask a surplus-sharing question raised, but not yet quantified, by ??: how the split of gains from trade between banks and firms, and the cross-subsidy from good to bad types within firms, vary with banks’ signal precision q .

Pooling and cross-subsidization at low signal precision. I simulate the model over the full admissible range of signal precision, $q \in [0.5, 1]$, holding all other parameters — π , $\bar{r}^g$, $\bar{r}^b$, μ_b , p_g , p_b — fixed at their rating-class-specific calibrated values (Table 4); $q = 0.5$ corresponds to an uninformative signal ($\alpha^{sg} = \alpha^{sb} = \mu_g$), and $q = 1$ to a perfect one ($\alpha^{sg} = 1$, $\alpha^{sb} = 0$). At $q = 0.5$, given the calibrated parameters, every rating class lies in a pooling equilibrium ($\phi^{si} > 0$): banks cannot condition on the signal and quote a common rate distribution to both types. Fig. 5 shows this directly — at $q = 0.5$ the expected rate paid by good and bad types is essentially indistinguishable in every rating class (e.g., 3.1205 versus 3.1209 for rating 6, 4.0719 versus 4.0725 for rating 7) — exactly the pooling outcome anticipated in ??: good types cross-subsidize bad types, paying, in expectation, a share of the surplus equal to $(1 - \pi)\mu_b(c_b - c_g)$, and banks do not manage to separate types.

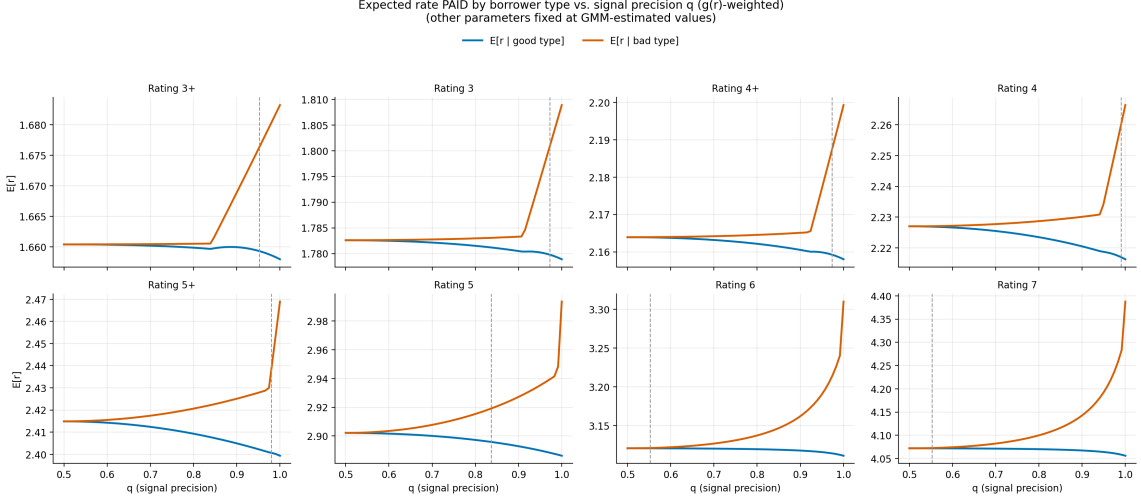


Figure 5: Expected interest rate paid by good vs. bad types as a function of signal precision q , by rating class. For each rating class, the panel holds π , $\bar{r}^g$, $\bar{r}^b$, μ_b , p_g , p_b fixed at their calibrated values (Table 4) and varies q over $[0.5, 1]$; the dashed vertical line marks the class’s own estimated $\hat{q}$. Rates are computed under the realized, transaction-weighted rate distribution $g(r)$ of Section 5 (Appendix D.1) — the rate actually paid by a borrower who obtains a loan, not the bank’s posted-quote distribution.

Rising rates for bad types and the shift to bad-only lending. As q increases, the signal grows more informative, the posterior belief α^{sb} that a bad-signal firm is actually good falls further, and conditioning on the signal becomes more valuable to the bank. Fig. 5 shows the rate paid by bad types rising, in every rating class, essentially monotonically in q , while the rate paid by good types is flat or mildly declining — so the type-conditional rate gap widens. This widening is concentrated late in the $[0.5, 1]$ range rather than gradual throughout: once q crosses the (calibration-specific) threshold at which ϕ^{si} turns negative, the bank abandons the pooling contract and lends to bad types only, sacrificing the good types whose offer now exceeds $\bar{r}^g$ in exchange for extracting more surplus from the bad types it continues to serve. Because this threshold depends on the calibrated parameters, its location differs sharply by rating class, mirroring the screening-intensity heterogeneity already documented in Section 5: for ratings 3+ through 5+, whose own estimated $\hat{q}$ is already close to 1 (Table 4), the model implies the bank is at, or close to, the bad-only region at its calibrated point, so for these classes the low- q region of Fig. 5 is more a counterfactual than a description of the estimated status quo — and q is, in any case, the least precisely pinned-down of the three calibrated parameters for these classes (Section 5.4). Ratings 6 and 7, in contrast, are estimated at $\hat{q} \approx 0.553$, indistinguishable from the uninformative floor of the domain, consistent with the near-zero screening intensity found in Section 5; for these two classes alone, Fig. 5 traces the full pooling-to-bad-only arc as an in-sample description rather than a counterfactual, with the two types paying statistically identical rates at $\hat{q}$ and separating only in the immediate vicinity of $q = 1$.

Full separation and profit extraction at $q = 1$. At $q = 1$ the signal is perfect and banks fully separate types, so the type-conditional rate gap in Fig. 5 is at its widest in every rating class (0.0252 for rating 3+, 0.1991 for rating 6, 0.3317 for rating 7). Fig. 7 shows the corresponding effect on the bank's bottom line: total expected profit per loan — the population-share-weighted average of the type-specific profits shown in Fig. 6 — is essentially flat while the bank pools, and rises once the bank starts separating types, reaching its maximum at $q = 1$ in every rating class. Acquiring a (near-)perfect signal therefore lets the bank extract the maximum profit from both types simultaneously; the cross-subsidy documented above disappears, at the cost, as anticipated in ??, of some good types exiting the market once their equilibrium offer exceeds $\bar{r}^q$.

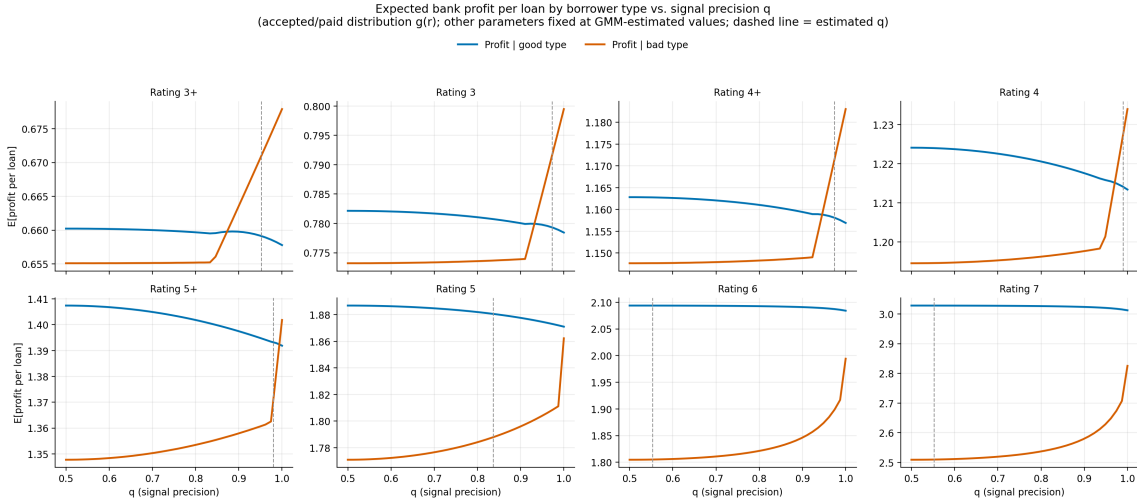


Figure 6: Expected bank profit per loan, by true borrower type, as a function of signal precision q , by rating class. Profit is $E[r \mid \text{type}] - c_{\text{type}}$ on the same realized-rate basis as Fig. 5; the dashed vertical line again marks the class's own estimated $\hat{q}$.

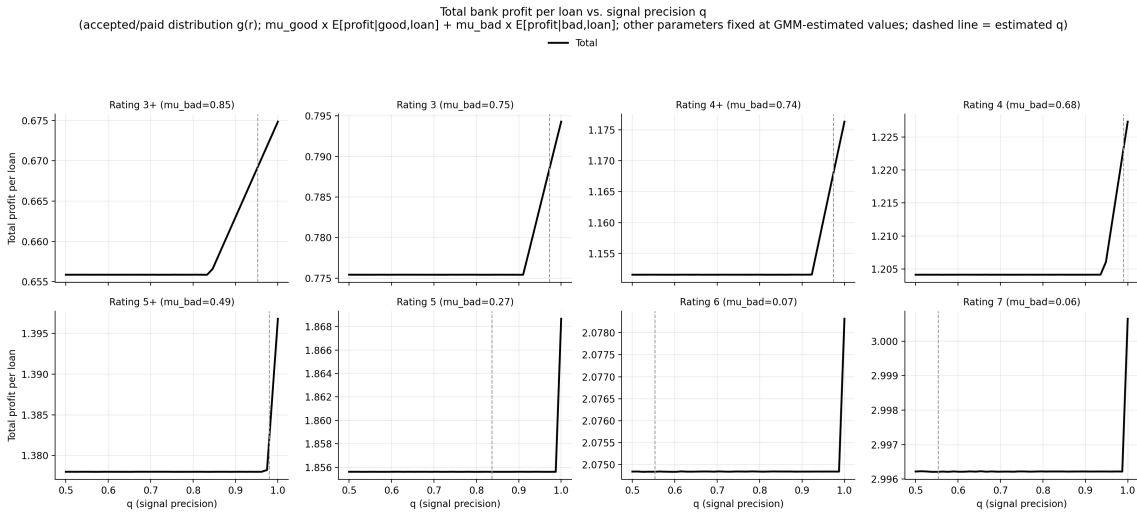


Figure 7: Total expected bank profit per loan as a function of signal precision q , by rating class: the population-share-weighted average $\mu_g E[\text{profit} \mid \text{good}] + \mu_b E[\text{profit} \mid \text{bad}]$ of the two series in Fig. 6. The dashed vertical line marks the class's own estimated $\hat{q}$.

Table 7: Redistribution from a perfect signal, baseline: percent of the rating class’s total surplus

Rating	Bad types	Banks	Good types
3+	-0.892	0.881	0.010
3	-0.847	0.833	0.013
4+	-0.806	0.786	0.020
4	-0.958	0.907	0.054
5+	-0.646	0.550	0.095
5	-0.438	0.274	0.163
6	-0.217	0.069	0.148
7	-0.308	0.071	0.236

Notes: change in each party’s expected surplus when signal precision moves from $q = 0.5$ to $q = 1$, as a percentage of the class’s maximum total surplus. Rows sum to zero: every class is in full pooling at $q = 0.5$, so no firm is rationed at either end and information redistributes without creating surplus.

Table 8: Redistribution from a perfect signal, baseline: percent of each party’s own surplus

Rating	Bad types	Banks	Good types
3+	-1.65	2.43	0.10
3	-1.80	2.19	0.08
4+	-1.82	1.93	0.13
4	-2.27	2.37	0.27
5+	-2.13	1.39	0.31
5	-2.76	0.65	0.39
6	-5.58	0.16	0.28
7	-7.75	0.16	0.46

Notes: the same changes, each divided by that party’s own surplus at $q = 0.5$ rather than by the class total. Rows do not sum. This is the denominator relevant to a bank weighing the cost of acquiring precision, and to a borrower judging the size of the repricing.

7 Conclusion

This paper asks how much of the dispersion in loan rates across otherwise similar firms reflects unobserved differences in borrower risk, and how much reflects market inefficiencies. Using French loan-level data, I document substantial rate dispersion among firms with similar credit risk and contract, even when they are clients of the same bank. I show that part of it reflects information acquired by the bank, as rate residuals carry forward information about the future evolution of firms’ creditworthiness, and this is strongest precisely where information is most valuable — for young and small firms, for longer-maturity loans, and during the financial crisis. The remainder is attributable to market inefficiencies that allow banks to depart from the law of one price.

To separate the two, I propose a novel equilibrium model of loan pricing in which dispersion is jointly determined by imperfect competition, modelled through search frictions, and a costly, imperfect public signal about observationally equivalent borrowers, whose precision is a feature of the market rather than a choice of the banks. Search frictions induce *within-type* dispersion, as banks' find it profitable to randomize to avoid being undercut by competitors, while pricing on the signal generates *between-type* dispersion, as banks charge on average higher rates to borrowers, together with the positive rate–default correlation I observe in the data. Because both forces are latent by construction, I recover them by calibrating the model to the variance of rate residuals, their correlation with default, and the maximum rate observed, separately for each Banque de France rating class.

The calibration delivers a clear quantitative message. The signal available in the market is highly precise, yet the quantitative role of information in accounting for observed dispersion is rather small: it accounts for at most about 5% relative to a no-information benchmark, and for essentially nothing in the riskiest classes. Imperfect competition is therefore the prevailing force behind the dispersion I measure, even where the market's information is close to perfect.

Information nonetheless matters for surplus sharing. Screening substantially benefits safer borrowers by limiting the two costs that adverse selection imposes on them: in a pooling equilibrium good types cross-subsidize bad types, and when adverse selection is severe they are priced above their reservation rate and exit the market altogether. More precise signals shrink both, lowering the borrowing costs of safer firms and reducing credit rationing, while bad types lose the subsidy they were implicitly receiving. A small aggregate contribution to dispersion and large consequences for how the surplus is shared are thus fully compatible. For policy, the two findings point in the same direction: since dispersion is driven mainly by imperfect competition rather than by risk-based pricing, a policy maker seeking to maximise firms' surplus and limit capital misallocation should aim at making the banking market more contestable, and at widening the information about borrowers that the market makes available, rather than at curbing banks' use of it.

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A Additional tables and figures

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LA COTE DE CRÉDIT (capacité de l'entreprise à honorer ses engagements financiers à 3 ans)	3++	Excellente
	3+	Très forte
	3	Forte
	4+	Assez forte
	4	Correcte
	5+	Assez faible
	5	Faible
	6	Très faible
	7	Appelant une attention spécifique présence d'au moins un incident de paiement significatif
	8	Menacée
	9	Compromise
	P	Procédure collective redressement ou liquidation judiciaire
	0	Pas de documentation comptable analysée et absence d'informations défavorables

Figure A.1: Source: Banque de France. Credit rating scale until 2022.

B Additional facts on rate dispersion

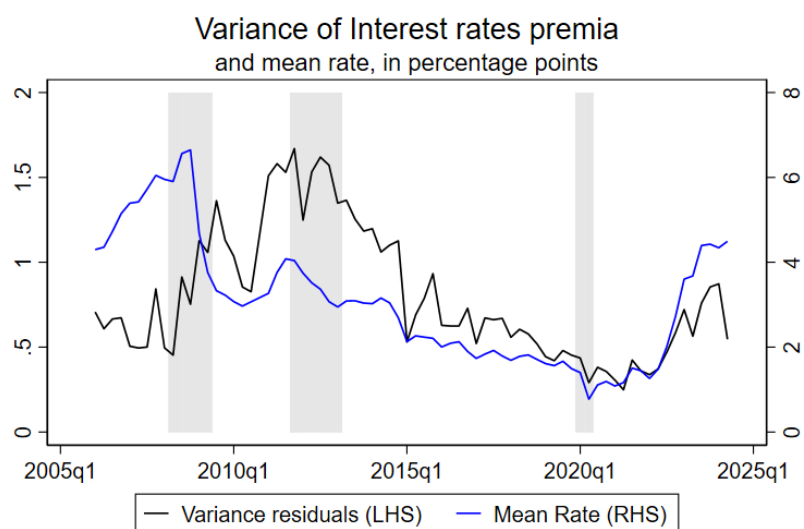
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Table B.1: Data: Nouveaux Credits (BdF), 2006q1-2024q2. Excludes credit lines and undefined maturity products. All credit and balance-sheet variables are expressed in terms of reported assets. Multiple products (dummy) equals one if the firm takes more than one product type in the quarter, counting the excluded products; Multiple loans, one product (dummy) equals one if it takes several loans in the quarter, all of the same type.

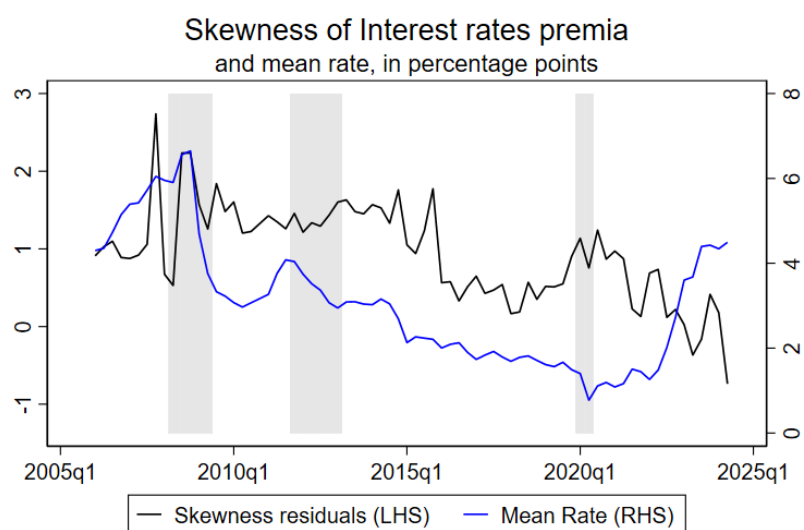
	Total effective rate					
log.Loan quantity (K-Eur)	-0.2762*** (0.000)	-0.1994*** (0.000)	-0.1969*** (0.000)	-0.1826*** (0.000)	-0.1859*** (0.000)	-0.1242*** (0.000)
Maturity (months)	-0.0006** (0.044)	0.0020*** (0.003)	0.0029*** (0.002)	0.0019*** (0.004)	0.0020*** (0.004)	0.0019*** (0.005)
Age	-0.0108*** (0.000)	-0.0076*** (0.000)	-0.0065*** (0.000)	-0.0057*** (0.000)	-0.0055*** (0.000)	-0.0033*** (0.000)
Age squared	0.0000*** (0.000)	0.0000*** (0.000)	0.0000*** (0.000)	0.0000*** (0.000)	0.0000*** (0.000)	0.0000*** (0.000)
Young firm 0-3y (dummy)	-0.0484*** (0.006)	-0.0482 (0.244)	-0.0539 (0.200)	-0.0288 (0.367)	-0.0259 (0.397)	-0.0654** (0.017)
New bank-firm relationship (dummy)	0.1549*** (0.000)	0.0027 (0.942)	-0.0044 (0.907)	-0.0317** (0.027)	-0.0321** (0.023)	-0.0455*** (0.003)
Variable rate (dummy)	0.3917*** (0.000)	0.4994*** (0.000)	0.2026** (0.022)	0.1477 (0.112)	0.1455 (0.112)	0.1603* (0.085)
Personal loan (dummy)	-0.3016** (0.044)	-0.0067 (0.958)	0.0758 (0.391)	0.0329 (0.686)	0.0374 (0.637)	0.0733 (0.309)
Individual entrepreneur (dummy)	0.4292*** (0.000)	0.4879*** (0.000)	0.4569*** (0.000)	0.3821*** (0.000)	0.3708*** (0.000)	0.2313*** (0.003)
Loan over assets	2.2482*** (0.000)	1.6201*** (0.000)	1.5301*** (0.000)	1.3391*** (0.000)	1.3166*** (0.000)	0.7238*** (0.001)
Pledged guarantees	1.6770*** (0.000)	0.2564 (0.117)	0.0107 (0.931)	-0.4193*** (0.000)	-0.3806*** (0.000)	-0.3332*** (0.000)
Revenues	-0.0940*** (0.000)	-0.0426*** (0.000)	-0.0342*** (0.001)	-0.0133*** (0.010)	-0.0113** (0.019)	-0.0116** (0.021)
Bank Debt	0.4944*** (0.000)	0.2907*** (0.000)	0.2538*** (0.001)	0.2188*** (0.000)	0.2368*** (0.000)	0.0177 (0.580)
Non-Bank Debt	0.0157 (0.782)	-0.0572 (0.729)	-0.0504 (0.773)	0.0206 (0.859)	0.0212 (0.848)	-0.0339 (0.665)
Mobilised credit	-0.2899*** (0.000)	-0.1095*** (0.007)	-0.0607 (0.125)	-0.0315 (0.377)	-0.0233 (0.483)	0.0185 (0.526)
Short Term Debt	0.3258*** (0.000)	0.0071 (0.952)	-0.0315 (0.803)	0.0779 (0.272)	0.0745 (0.282)	-0.1557*** (0.004)
Undrawn credit lines	-0.0385 (0.762)	-0.0994 (0.410)	-0.1246 (0.329)	-0.2140** (0.013)	-0.1931** (0.025)	-0.1339* (0.085)
Multiple products (dummy)	0.0296 (0.122)	0.0339 (0.120)	0.0025 (0.897)	-0.0024 (0.869)	-0.0012 (0.933)	0.0196* (0.055)
Multiple loans, one product (dummy)	-0.1562** (0.017)	-0.0357 (0.506)	-0.0565 (0.285)	-0.0224 (0.471)	-0.0227 (0.459)	0.0029 (0.909)
Quarter FE	✓					
Lender × Quarter FE		✓				
Lender × Quarter × Loan Type FE			✓			
Lender × Quarter × Loan Type × 2digit Sector FE				✓	✓	✓
Region FE					✓	✓
Group indicator, Rating FE						✓
Observations	449,973	449,973	449,973	449,973	449,973	449,973
R ²	0.57	0.72	0.76	0.82	0.82	0.83

p-values in parentheses

* p < 0.10, ** p < 0.05, *** p < 0.01

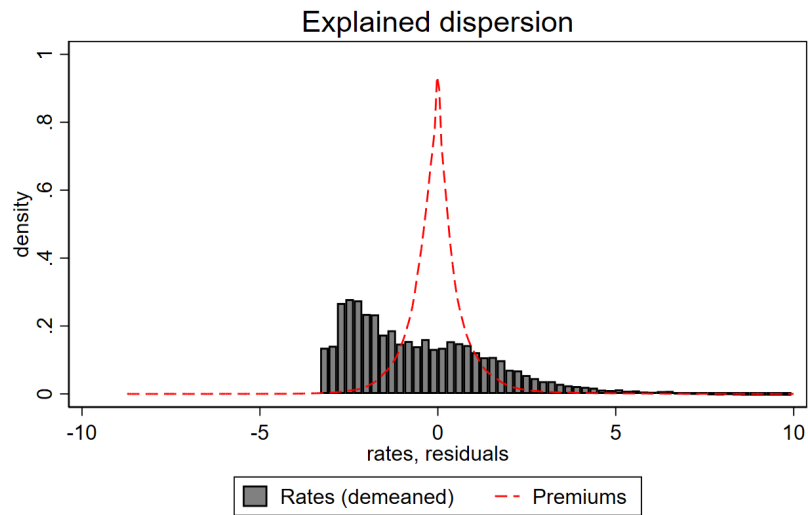


Source: Nouveaux Crédits, BdF, 2006q1-2024q2. All term loans.
 Recession dates for France from AFSE: 2008q2-2009q2, 2011q4-2013q1, 2020q1-2020q2

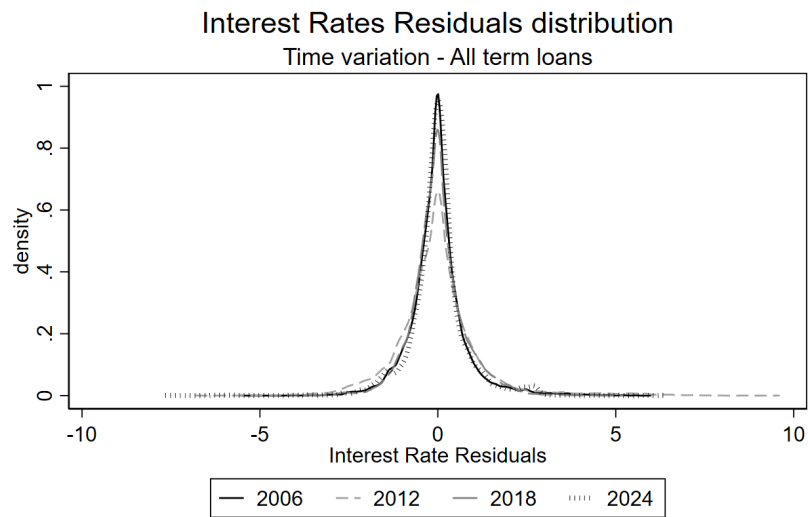


Source: Nouveaux Crédits, BdF, 2006q1-2024q2. All term loans.
 Recession dates for France from AFSE: 2008q2-2009q2, 2011q4-2013q1, 2020q1-2020q2

Figure B.2: Variance (top) and skewness (bottom) of interest rate residuals by quarter, black line, left axis, against the mean interest rate, blue line, right axis, both in percentage points. Shaded bands are French recessions (AFSE): 2008q2–2009q2, 2011q4–2013q1 and 2020q1–2020q2. Data: Nouveaux Crédits (BdF), 2006q1–2024q2, all term loans.



Source: Nouveaux Credits, BdF, 2006q1-2024q2
Excludes credit lines and personal loans.



Source: Nouveaux Credits, BdF, 2006q1-2024q2
Excludes credit lines and personal loans.

Figure B.3: Regression outcome. Top: residuals versus actual rates, all sample. Bottom: residuals distribution for selected years.

B.1 Additional specification 1: Firm fixed effects

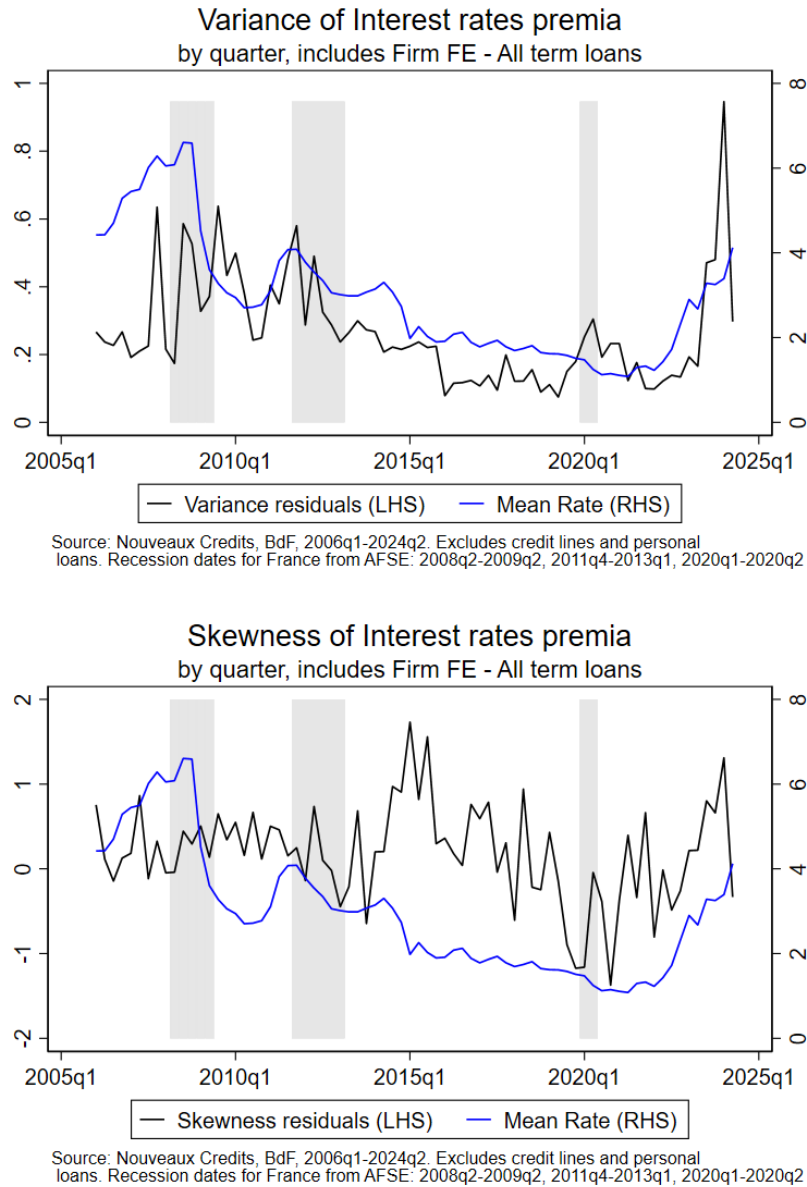
	Total effective rate					
Loan quantity (K-Eur)	-0.0000*** (0.000)	-0.0000*** (0.006)	-0.0000 (0.486)	-0.0000 (0.143)	-0.0000** (0.044)	-0.0000 (0.117)
Maturity (months)	-0.0016*** (0.000)	0.0013*** (0.000)	0.0015*** (0.000)	0.0015*** (0.000)	0.0014*** (0.000)	0.0014*** (0.000)
Variable rate (dummy)	0.2381*** (0.000)	0.1357*** (0.000)	0.2176*** (0.000)	0.2368*** (0.000)	0.2113*** (0.000)	0.1995*** (0.000)
Loan over assets	0.0524*** (0.000)	-0.0068** (0.025)	-0.0076*** (0.009)	-0.0070** (0.012)	-0.0069** (0.012)	-0.0074*** (0.007)
Quarter FE	✓					
Firm × Quarter FE		✓	✓	✓	✓	
Lender FE			✓			
Lender × Quarter FE				✓	✓	
Firm × Lender × Quarter FE						✓
Loan Type FE					✓	✓
Observations	590,621	197,343	197,328	196,048	196,042	156,211
R^2	0.51	0.92	0.93	0.94	0.94	0.94

p-values in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B.2: "Data: Nouveaux Credits (BdF), 2006q1-2024q2" Excludes credit lines and personal loans. All credit and " " balancesheet variables are expressed in terms of reported assets."

Figure B.4: variance and skewness of residuals from firm FE specification



B.2 Additional specification 2: truncated regression approach [Back to main ↑](#)

One possible concern is that it is not possible to observe rates below 0. Regression approaches assuming normality may be downward biased, predicting negative rates even when they are impossible to observe. Two approaches can help address this issue. Firstly, one can verify to what extent the OLS is predicting negative values. It seems that most of the variation being absorbed by fixed effect, this is a minor concern, as 0.4% of predictions fall slightly below zero. Secondly, the same analysis can be repeated using Tobit regression, which accounts for this lower bound; this approach is more limited towards the inclusion of many interactions of fixed effects, thus it forces a more parsimonious approach. In any case, the resulting distribution of residuals does not seem very affected.

B.3 Rate dispersion and bank heterogeneity

One question that naturally arises is: is this dispersion hiding heterogeneity across banks? The answer is yes. There is consistent heterogeneity between banks in the explained variation, as shown below by repeating the above estimation of rates by bank, and collecting the R^2 .

$$\text{Rate}_{i,b,t} = \alpha + \beta \cdot \text{LoanChar}_{i,b,t} + \gamma \cdot \text{FirmChar}_{i,t} + \lambda_t + \varepsilon_{i,b,t} \quad (8)$$

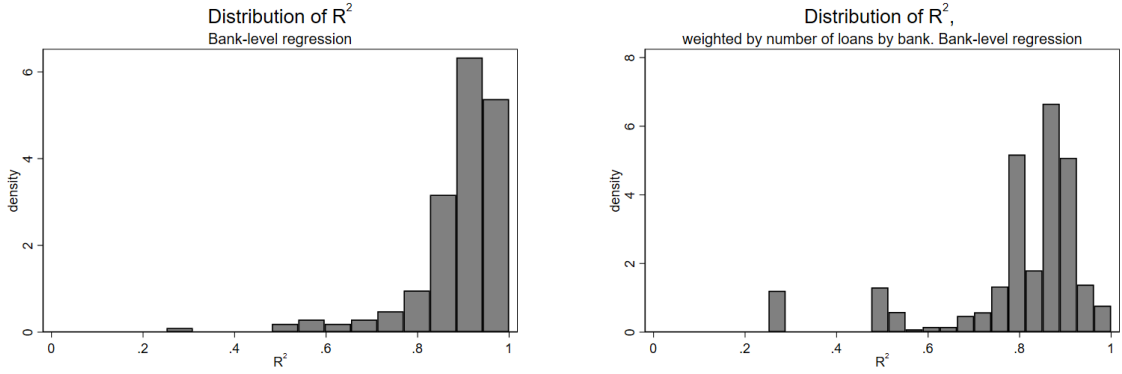


Figure B.5: The right image reports the distribution of R^2 weighted by the number of loans issued.

In the following section the credit rating of firms will be used as a source of public information available to banks and to study the evolution of firm's outcome in terms of their default probability/outcome. Consequently, it is interesting at this stage to report to which extent banks use these ratings. Firstly, the same estimation is repeated by including only the rating as explanatory variable, along with time and loan type FE and by lender, and collect the R^2 .

$$\text{Rate}_{i,b,t} = \alpha + \beta \cdot \text{Firm Rating}_{i,b,t} + \lambda_t + \varepsilon_{i,b,t} \quad (9)$$

where the only fixed effects kept are by loan type and quarter.

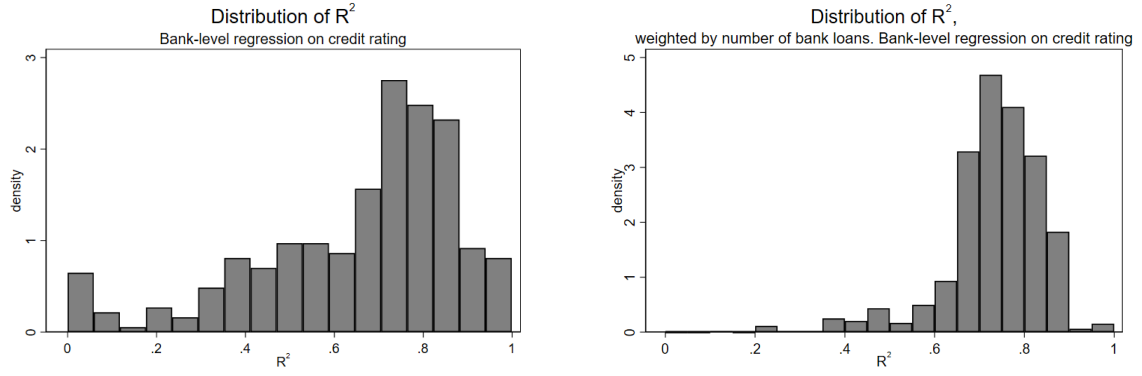


Figure B.6: The right image reports the distribution of R^2 weighted by the number of loans issued.

Secondly, one can look at how much of the variation is explained when including the public rating, in particular by collecting the increment in the R^2 .

$$\begin{aligned}
 \text{Rate}_{i,b,t} &= \alpha + \lambda_t + \varepsilon_{1,i,b,t} && \rightarrow \text{Get } R_1^2 \\
 \text{Rate}_{i,b,t} &= \alpha + \beta \cdot \text{Firm Rating}_{i,b,t} + \lambda_t + \varepsilon_{2,i,b,t} && \rightarrow \text{Get } R_2^2 \\
 \Delta R^2 &= R_2^2 - R_1^2
 \end{aligned}$$

where the only fixed effects kept are by loan type and quarter.

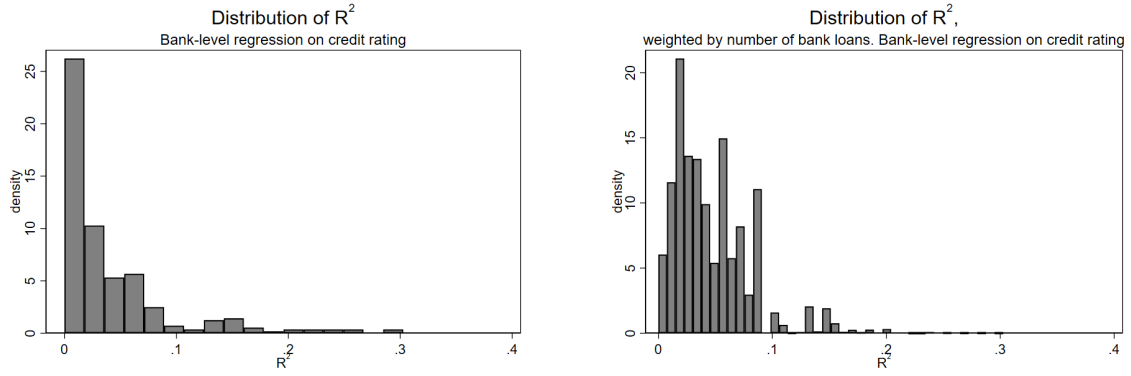


Figure B.7: The right image reports the distribution of R^2 weighted by the number of loans issued.

B.4 Residuals and market competition

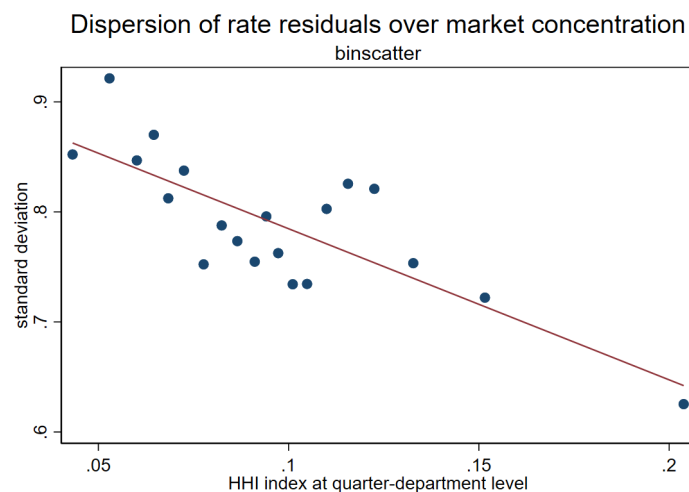
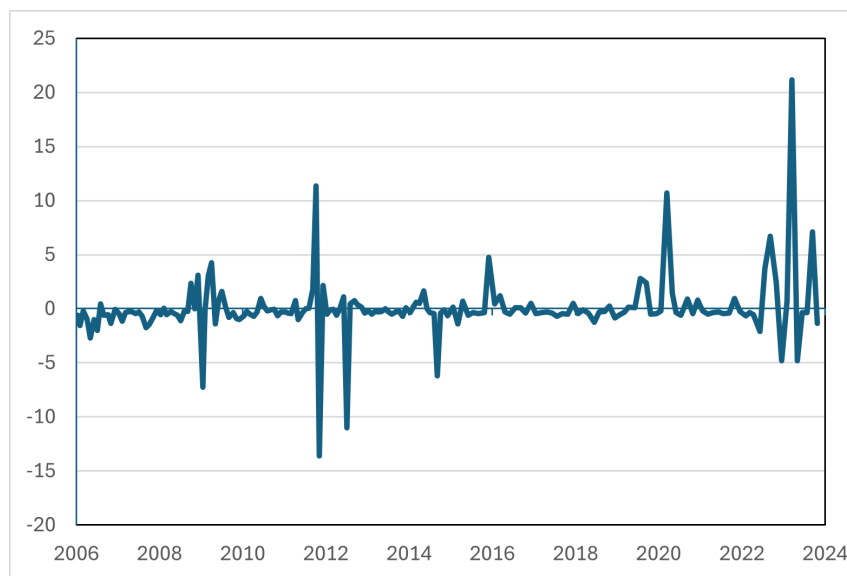


Figure B.8: Unexplained rate residuals' dispersion over HHI by department-quarter.

B.5 Monetary policy implications

Figure B.9: Interest rate shock factor estimated from changes in short term OIS rates in a narrow window after the ECB press release.



B.6 Alternative specifications: Rate residuals and changes in default probability

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This section reports four alternatives to Eq. (2). The first replaces the rating revision with the default itself. The other three turn the regression around and put the revision in default probability on the left, so that the coefficient of interest is the one on the

rate r_{ilbt} itself. The reversal changes the normalisation, not the question: a positive coefficient says that firms charged a higher rate at issuance are revised towards a higher default probability afterwards, which is what $\beta_k > 0$ says in Eq. (2). Magnitudes are not comparable across the two directions — one is measured in percentage points of rate per percentage point of default probability, the other the other way round.

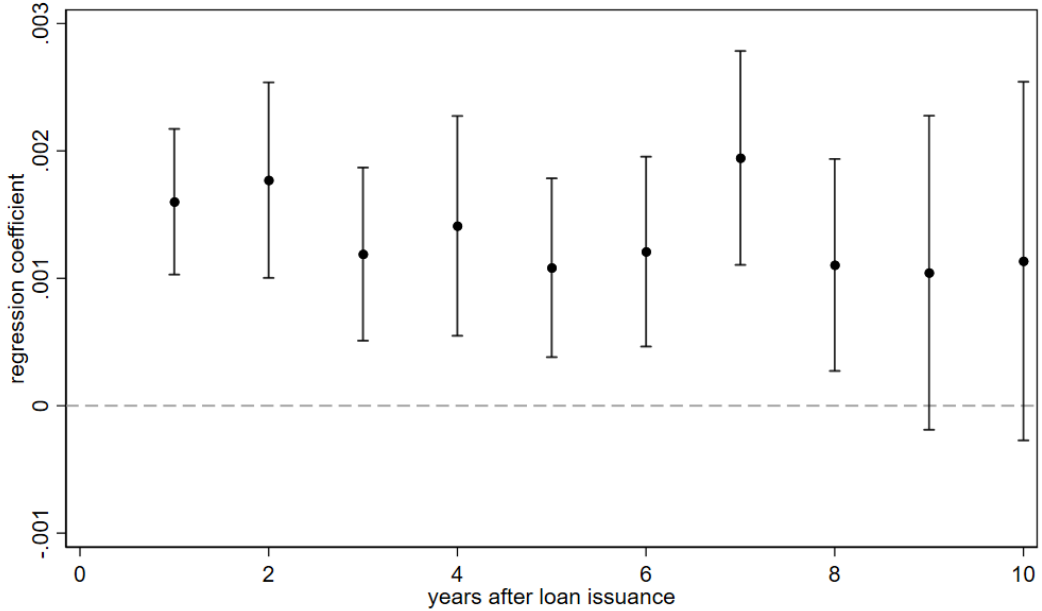
B.6.1 Actual defaults

I repeat the test replacing the rating revision with the default itself; for every k year, I run:

$$r_{ilbt} = \alpha + \sum_{j=1}^k \beta_j^D D_{i,t+j} + \mathbf{x}_{ilbt}' \gamma + \mu_{bt\tau s} + \varepsilon_{ilbt}, \quad (10)$$

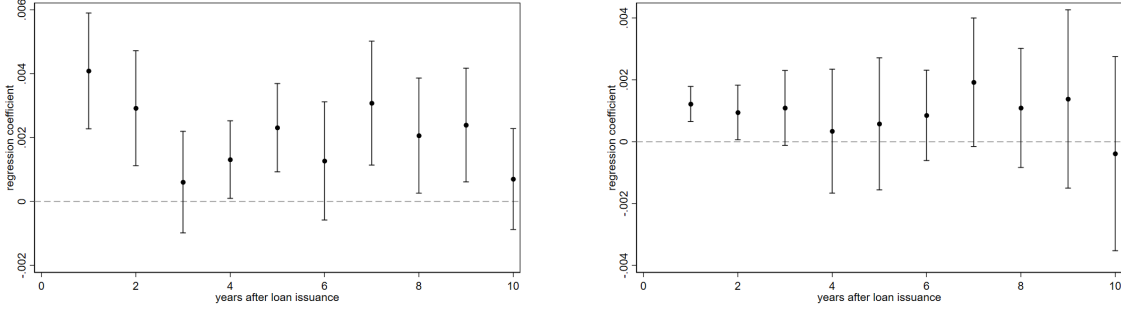
in which $D_{i,t+j}$ equals 100 if firm i defaults in the j -th year after issuance and 0 otherwise. This test uses the annual rating file, which records a default whenever it occurs, so it needs no balanced panel; it adds a dummy for a new bank-firm relationship to $\mathbf{x}_{ilbt}$ and clusters standard errors by lender and by quarter. I estimate Eq. (10) once per horizon $k = 1, \dots, 10$, each regression carrying the default indicators up to that horizon, and report β_k^D .

Figure B.10: Regression coefficient of actual default, controlling for lags and observables, on interest rates.



Rating revisions on the left-hand side. Three choices distinguish the three specifications that follow: whether the revision is the year-on-year change or the cumulative change since issuance, whether the earlier revisions enter as controls, and whether the sample is balanced across horizons. Everything else is held fixed. All three carry the same controls $\mathbf{x}_{ilbt}$ and fixed effects $\mu_{bt\tau s}$ as Eq. (2), cluster standard errors by lender,

Figure B.11: Correlation computed separately for firms with a good ex ante rating, classes 3++ to 4+ (left), and a bad one, classes 5 to 9 (right). Classes 4 and 5+ are left out.



drop loans to firms already in default at issuance ($P_{i,t} \neq 1$), and run one regression per horizon $k = 1, \dots, 10$, reporting the coefficient on the rate. The rating-class split is the one used in Fig. B.18: classes 3++ to 4+ for the good group and classes 5 to 9 for the bad one, with classes 4 and 5+ left out.

B.6.2 Year-on-year revisions, past revisions controlled for, balanced sample

For each horizon k I estimate

$$\Delta P_{i,t+k} = \alpha_k + \theta_k r_{ilbt} + \sum_{j=1}^{k-1} \delta_{kj} \Delta P_{i,t+j} + \mathbf{x}'_{ilbt} \gamma_k + \mu_{bt\tau s} + \varepsilon_{ilbt}, \quad (11)$$

in which $\Delta P_{i,t+k}$ is the change in default probability between years $k - 1$ and k after issuance, in percentage points, as in Eq. (2). The sample is balanced: a loan enters only if every revision $\Delta P_{i,t+1}, \dots, \Delta P_{i,t+10}$ is observed, so the same firms identify every θ_k . This is the closest counterpart to Eq. (2), which is also estimated on the balanced panel. The rate predicts the first-year revision, $\theta_1 = 0.064$ (s.e. 0.017), and the coefficient falls towards zero by the tenth year. The pattern holds for the good rating classes and is too noisy to sign for the bad ones.

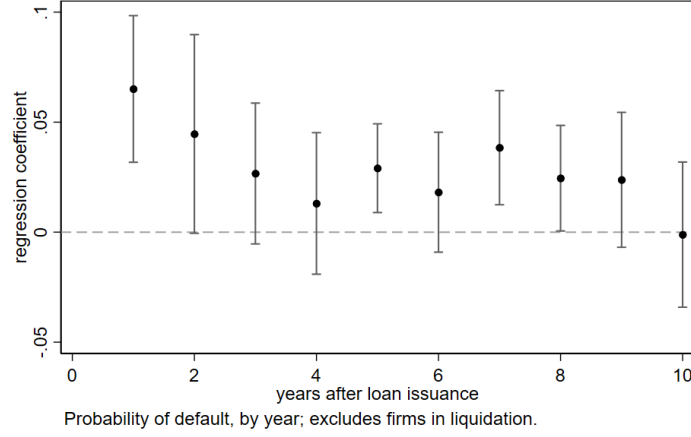
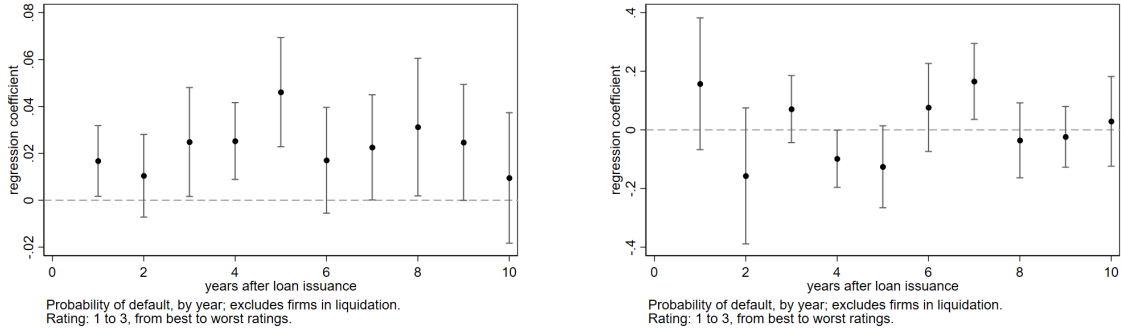


Figure B.12: Coefficient θ_k on the loan rate in Eq. (11), by horizon k . The dependent variable is the year-on-year change in the firm's default probability between years $k - 1$ and k after issuance, in percentage points. Balanced sample; past revisions controlled for; bars are 95% confidence intervals, standard errors clustered by lender.

Figure B.13: Coefficient θ_k in Eq. (11), estimated separately for firms with a good ex ante rating, classes 3++ to 4+ (left), and a bad one, classes 5 to 9 (right). Classes 4 and 5+ are left out. Note the different vertical scales.



B.6.3 Year-on-year revisions, past revisions controlled for, unbalanced sample

Eq. (11) keeps only firms followed for the full ten years, which selects on survival in the rating file. Dropping that requirement, I estimate the same equation horizon by horizon on whatever firms are observed at that horizon:

$$\Delta P_{i,t+k} = \alpha_k + \theta_k r_{ilbt} + \sum_{j=1}^{k-1} \delta_{kj} \Delta P_{i,t+j} + \mathbf{x}'_{ilbt} \gamma_k + \mu_{bt\tau s} + \varepsilon_{ilbt}, \quad (12)$$

so the sample shrinks with k and differs across horizons. The coefficients are larger than on the balanced panel — $\theta_1 = 0.14$ (s.e. 0.022) — and decline to zero by the tenth year. The gap between the two samples says that the firms that stay in the rating file for ten years are the ones whose rates carry less information about their own revisions.

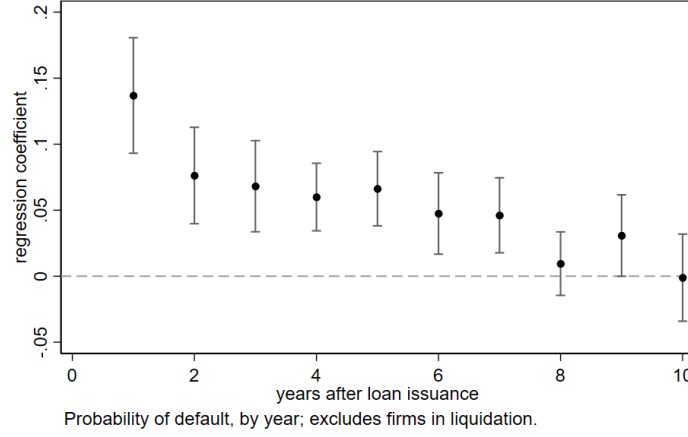
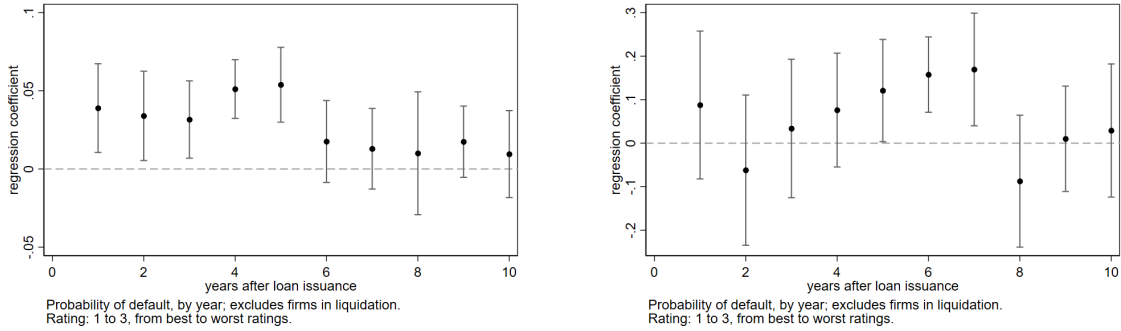


Figure B.14: Coefficient θ_k on the loan rate in Eq. (12), by horizon k . The dependent variable is the year-on-year change in the firm's default probability between years $k - 1$ and k after issuance, in percentage points. Unbalanced sample; past revisions controlled for; bars are 95% confidence intervals, standard errors clustered by lender.

Figure B.15: Coefficient θ_k in Eq. (12), estimated separately for firms with a good ex ante rating, classes 3++ to 4+ (left), and a bad one, classes 5 to 9 (right). Classes 4 and 5+ are left out. Note the different vertical scales.



B.6.4 Cumulative revision since issuance

The year-on-year revision at horizon k is an increment, so θ_k in Eqs. (11) and (12) measures how much of the revision arriving in year k alone the rate anticipates. Measuring the revision from issuance instead asks how much of the firm's total drift in default probability the rate anticipates by year k :

$$\Delta P_{i,t \rightarrow t+k} = \alpha_k + \theta_k r_{ilbt} + \mathbf{x}'_{ilbt} \gamma_k + \mu_{bt\tau s} + \varepsilon_{ilbt}, \quad (13)$$

in which $\Delta P_{i,t \rightarrow t+k} = 100 \times (P_{i,t+k} - P_{i,t})$, the same cumulative object used for the demographic splits in Fig. 4. Past revisions no longer enter, since the dependent variable already contains them. At $k = 1$ the two measures coincide and Eq. (13) reproduces Eq. (12), $\theta_1 = 0.14$. Beyond that the coefficient stays positive at every horizon, peaking at $\theta_5 = 0.18$ (s.e. 0.023) and still 0.11 at ten years. The rate anticipates a level shift in

the firm's default probability that does not unwind, and the decline in Fig. B.14 reflects the exhaustion of new information, not a reversal of the old.

Splitting by rating class sharpens the reading. For the good classes the cumulative coefficient rises from 0.039 at one year to 0.14 at five and is positive at every horizon. For the bad classes it is two to three times larger at the longer horizons but is not distinguishable from zero in the first three years: rates predict where bad borrowers end up, not when they get there.

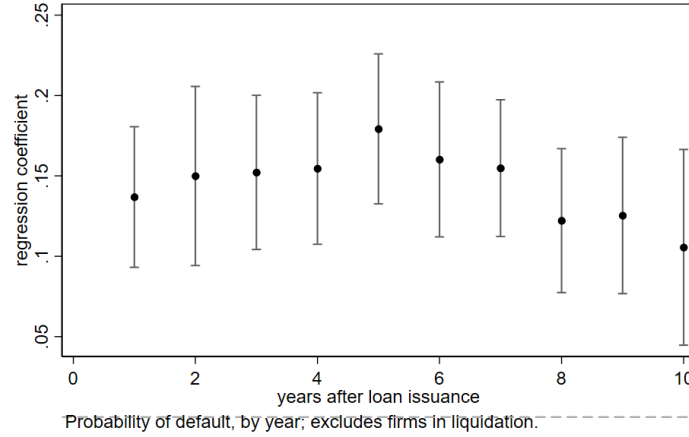


Figure B.16: Coefficient θ_k on the loan rate in Eq. (13), by horizon k . The dependent variable is the cumulative change in the firm's default probability between issuance and year k , in percentage points. Bars are 95% confidence intervals, standard errors clustered by lender.

Figure B.17: Coefficient θ_k in Eq. (13), estimated separately for firms with a good ex ante rating, classes 3++ to 4+ (left), and a bad one, classes 5 to 9 (right). Classes 4 and 5+ are left out. The horizontal axis is the horizon k of the cumulative revision. Note the different vertical scales.

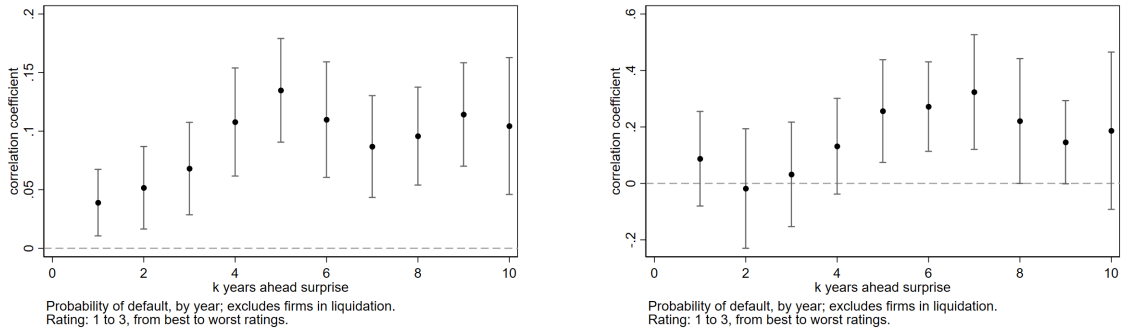


Figure B.18: Correlation computed separately for firms with a good ex ante rating, classes 3++ to 4+ (left), and a bad one, classes 5 to 9 (right). Classes 4 and 5+ are left out.

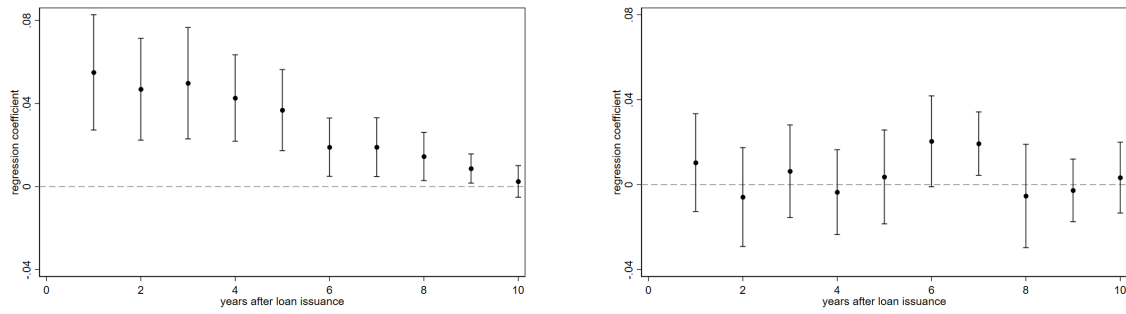
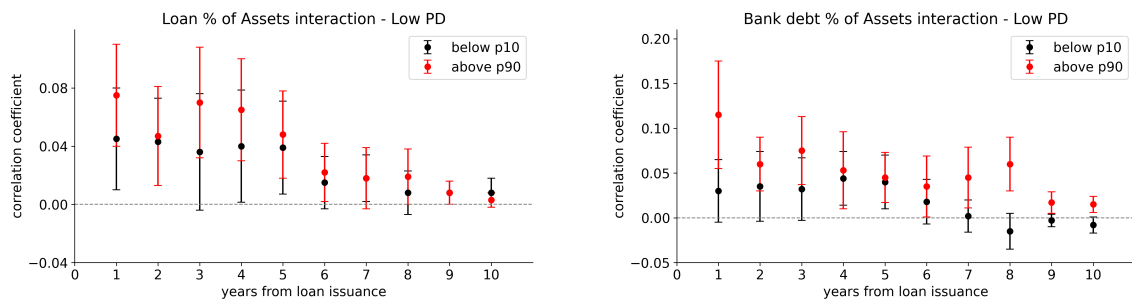


Figure B.19: Interactions of ex-post default probability and loan-size over assets (left) and bank debt over assets (right)



The figure compares the correlation coefficients of residual rates and unexpected changes in ex-post default probability across percentiles of loan-size over assets (left) and size of bank debt over assets (right) for firms with a good ex ante rating (classes 3++ to 4+).

B.7 Correlation across firms' demographics

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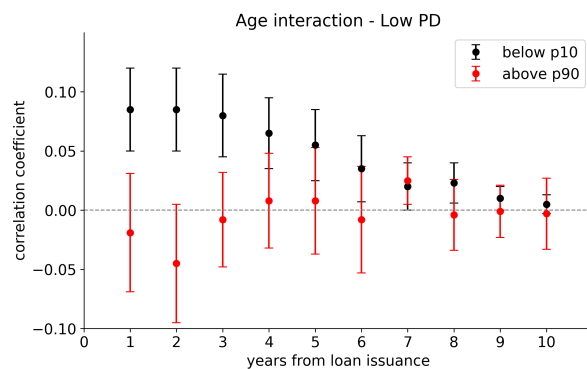


Figure B.20: The figure shows the correlation coefficients of the interactions of ex-post default probability and age (below 10th vs above 90th age percentile). Sample of firms with a good ex ante rating (classes 3++ to 4+).

B.7.1 Fixed vs variable rate test

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	Default Probability						
Variable rate (dummy)	0.1176** (0.014)		0.1407 (0.127)	0.1039** (0.026)			
cum dRate end y2		0.0433 (0.229)			0.0374 (0.254)		
annual dRate end y1			0.0812 (0.600)		0.0524 (0.711)	0.0425 (0.773)	
annual dRate end y2			0.0567 (0.330)		0.0363 (0.551)	0.0128 (0.879)	
Variable rate (dummy) $\times$ Loan over assets				0.1932 (0.370)			
cum dRate end y2 $\times$ Loan over assets					0.0711 (0.264)		
annual dRate end y1 $\times$ Loan over assets					-0.1330 (0.763)		
annual dRate end y2 $\times$ Loan over assets					0.0905 (0.148)		
annual dRate end y1 $\times$ Bank Debt share						0.0058 (0.980)	
annual dRate end y2 $\times$ Bank Debt share						0.0986 (0.623)	
Loan over assets	0.2250** (0.030)	0.2002** (0.035)	0.2048** (0.035)	0.1808* (0.094)	0.1894* (0.054)	0.1849* (0.059)	0.1929* (0.054)
Bank Debt share	0.0846** (0.040)	0.1139** (0.019)	0.1134** (0.019)	0.0850** (0.034)	0.1118** (0.022)	0.1114** (0.023)	0.1127** (0.025)
Lender $\times$ Month $\times$ Loan Type $\times$ 2digit Sector FE	✓	✓	✓	✓	✓	✓	✓
Region FE	✓	✓	✓	✓	✓	✓	✓
All other observables	✓	✓	✓	✓	✓	✓	✓
Observations	102,463	96,414	96,414	102,463	96,414	96,414	96,414
R^2	0.42	0.40	0.40	0.42	0.40	0.40	0.40

p-values in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B.3: Data: Nouveaux Crédits (BdF), 2006q1-2024q2. Term loans, at least 30-month maturity, only firms with a rating. Dependent variable: change_PD3. Exclude loans with non-specified reference rate.

Baseline estimates. I restrict the baseline to loans with a maturity of at least 30 months. The restriction keeps the loan outstanding across the three-year window over which I measure ΔPD , so the estimate is not driven by loans refinanced before a rate increase can pass through. Table B.3 reports the estimates. I find no significant impact of increased debt-servicing costs on rating worsening, suggesting that, at least in my sample, an increase in rates inducing an increase in leverage cannot alone induce a worsening firm-level risk. Interestingly, while the rate increase does not induce a worse

ex-post outcome, the loan interest rate remains a significant predictor of ex-post riskiness.

Identifying assumptions. The exercise rests on three assumptions.

- (i) *Exogeneity.* Reference-rate movements are set by monetary policy and aggregate conditions, not by firm i 's own prospects. This is what the within-cell comparison buys: β is identified off two loans made by the same lender, in the same month, of the same type, to firms in the same industry, that differ in reference-rate exposure. The assumption fails only if a loan's index is chosen in anticipation of that firm's own trajectory.
- (ii) *No sorting on private information or rate expectations.* Conditional on $\mathbf{x}_{ilbt}$ and $\mu_{bt\tau s}$, firms do not choose between fixed and floating on information the econometrician cannot observe — including a view on where rates are heading.
- (iii) *Parallel trends.* Absent the rate increase, default probabilities of fixed- and floating-rate borrowers would have evolved in the same way.

The rest of this section gives evidence on (ii) and (iii), and then repeats the exercise on the sample for which (ii) is least likely to bind.

Self-selection and parallel trends. I mitigate the concern behind (ii) by showing that the contract choice is predominantly determined by supply-side lender preferences rather than firm-specific unobservables. Focusing on firms with a rating at the time of the loan, I estimate a linear probability model of the form:

$$Floating_{loan} = \beta X^{observables} + \delta_{t,b,p,s} + \varepsilon_{loan} \quad (14)$$

where $Floating_{loan}$ is a dummy for variable-rate loans and $\delta_{t,b,p,s}$ is the fixed effects interaction term for month, bank, loan purpose and firm sector. Following a variance decomposition approach, I gradually saturate the model with fixed effects. As shown in Table B.4, the lender-month dimension alone accounts for approximately 60% of the variation, with the full specification explaining up to 91% of the total variance. This suggests that the contract choice is predominantly driven by supply-side and sectoral factors rather than idiosyncratic firm risk; in fact, banks may prefer to offer floating rate contracts if a variable rate schedule is applied to their own liabilities (see Kirti (2020)).

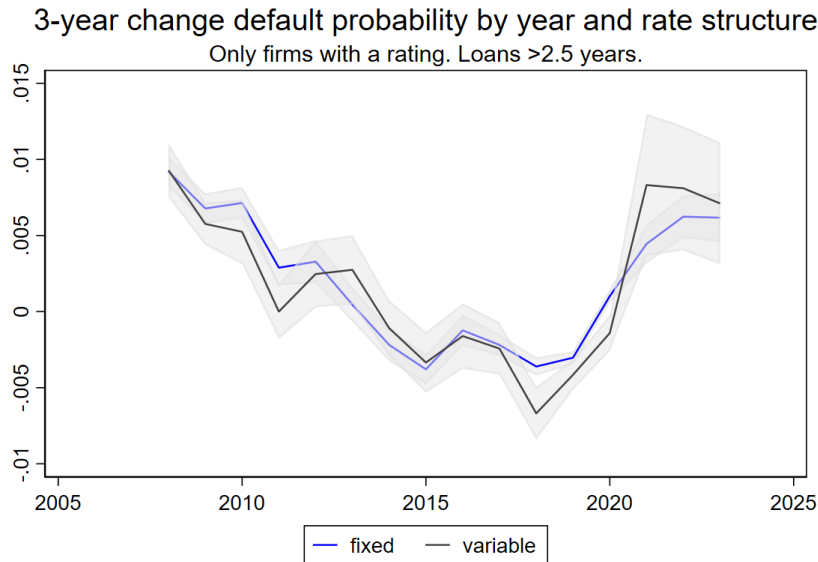
I also check the balance of observable characteristics between firms taking fixed and variable rate loans. While I find that the two groups differ in some characteristics (variable: fewer firms, larger size, older, riskier, larger financial debt), I include all those characteristics as controls in the following regression, to mitigate the issue of selection

Table B.4: Variance Decomposition of Variable Rate Choice

Fixed Effects Included	Adjusted R^2
Month	0.04
+ Lender	0.63
+ Loan Type	0.86
+ Firm Sector	0.91

bias. I also check the parallel trends assumption by plotting the evolution of rating transitions for firms taking fixed vs variable rate loans. I find that rating transitions for firms taking variable rate loans do not differ significantly from those taking fixed rate loans during the period considered. Fig. B.21 plots the two series.

Figure B.21: Mean three-year change in the default probability by year, for fixed- and floating-rate borrowers separately. Term loans over 2.5 years, firms with a rating. Shaded: 95% confidence bands.



This evidence overall mitigates concerns about selection bias.

The sharpest test: loans issued in 2021q3–q4. Assumption (ii) binds least where firms cannot have been acting on a view about future rates. I therefore repeat the exercise on loans issued in the second half of 2021, at the end of the ZLB period: the policy rate was still at its floor, and the tightening that followed in 2022 and 2023 came as a surprise. A firm choosing between a fixed and a floating contract in those two quarters was not choosing against an anticipated increase, so sorting on rate expectations is ruled out by the timing of the sample rather than by controls. This is the version of the exercise in which the leverage channel, if it operated, would be hardest to hide. Here I lower the maturity cut to 24 months: the 2021q3–q4 window leaves too few loans at the 30-month threshold for the estimator to report standard errors on several coefficients. Table B.5

reports the estimates. I again find no significant effect of reference rate increases on rating worsening.

	Default Probability - 2021q3-2021q4 only						
Variable rate (dummy)	-0.5698 (0.169)	-6.1935 (0.193)	-0.5121 (0.186)				
annual dRate end y2		1.8860 (0.261)		-0.3828* (0.095)	-0.4445 (0.218)		
Variable rate (dummy) $\times$ Loan over assets			-0.6271 (0.249)				
annual dRate end y2 $\times$ Loan over assets				-0.3410 (0.228)			
annual dRate end y2 $\times$ Bank Debt share						0.0555 (0.915)	
Loan over assets	-0.5329 (0.374)	-1.1142 (0.168)	-1.1243 (0.216)	-0.1584 (0.740)	-0.6316 (0.224)	-0.6318 (0.223)	-1.1395 (0.177)
Bank Debt share	0.0986 (0.725)	-0.2614 (0.316)	-0.2672 (0.448)	0.1401 (0.631)	-0.2234 (0.375)	-0.2234 (0.375)	-0.2780 (0.364)
Lender $\times$ Month $\times$ Loan Type $\times$ 2digit Sector FE	✓	✓	✓	✓	✓	✓	✓
Region FE	✓	✓	✓	✓	✓	✓	✓
All other observables	✓	✓	✓	✓	✓	✓	✓
Observations	4,324	4,089	4,089	4,324	4,089	4,089	4,089
R^2	0.41	0.41	0.41	0.41	0.41	0.41	0.41

p-values in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B.5: Data: Nouveaux Crédits (BdF), 2006q1-2024q2. Term loans, maturity at least 24 months, only firms with a rating. Dependent variable: change_PD3. Exclude loans with non-specified reference rate.

Including the 2023 rate increase. The exercise above uses loans issued in 2021q3–q4 and the 2022 rate increase alone. Restricting the increase to 2022 keeps the horizon close to the two years over which the rating is revised, but it also discards the second leg of the tightening. Table B.6 repeats the exercise on the same sample, letting the reference rate change accumulate through 2023 as well. The coefficients on the rate increase and on its interactions with loan-over-assets and bank debt share remain insignificant, so the conclusion in the main text does not depend on where the rate window is cut.

	Default Probability - 2021q3-2021q4 only						
Variable rate (dummy)	-0.5698 (0.169)	-19.6243 (0.479)	-0.5121 (0.186)				
annual dRate end y2		5.0816 (0.477)		0.0935 (0.943)	0.6439 (0.316)		
annual dRate end y3		2.6790 (0.563)		-0.7103 (0.718)	-1.5704 (0.179)		
Variable rate (dummy) $\times$ Loan over assets			-0.6271 (0.249)				
annual dRate end y2 $\times$ Loan over assets				5.0753 (0.781)			
annual dRate end y2 $\times$ Bank Debt share					-0.6506 (0.424)		
annual dRate end y3 $\times$ Loan over assets				-7.6615 (0.769)			
annual dRate end y3 $\times$ Bank Debt share					0.9643 (0.155)		
Loan over assets	-0.5329 (0.374)	-1.1153 (0.168)	-1.1298 (0.233)	-0.1584 (0.740)	-0.6351 (0.216)	-0.6550 (0.287)	-1.1379 (0.283)
Bank Debt share	0.0986 (0.725)	-0.2617 (0.313)	-0.2685 (0.569)	0.1401 (0.631)	-0.2242 (0.369)	-0.2431 (0.431)	-0.2795 (0.674)
Lender $\times$ Month $\times$ Loan Type $\times$ 2digit Sector FE	✓	✓	✓	✓	✓	✓	✓
Region FE	✓	✓	✓	✓	✓	✓	✓
All other observables	✓	✓	✓	✓	✓	✓	✓
Observations	4,324	4,089	4,089	4,324	4,089	4,089	4,089
R^2	0.41	0.41	0.41	0.41	0.41	0.41	0.41

p-values in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B.6: Data: Nouveaux Crédits (BdF), 2006q1-2024q2. Term loans, maturity at least 24 months, only firms with a rating. Dependent variable: change_PD3. Exclude loans with non-specified reference rate.

C Model appendix

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C.1 Equilibrium under an informative signal

I now examine the scenario where banks can acquire an imperfect signal about each firm's type for a fixed cost C_s . This allows me to analyze the endogenous information acquisition decision and characterize the equilibrium as signal precision varies from uninformative to perfect.

Signal-Specific Equilibria. Conditional on acquiring the signal, banks face σ_g firms

with good signals and σ_b firms with bad signals. For each group, the bank solves a rate-setting problem analogous to the no-information case, but with updated beliefs about the shares of types in the population receiving signal s_i . The sign of ϕ^{s_i} determines whether the equilibrium is pooling or bad-only within each signal group. For each signal s_i , define:

$$\phi^{s_i} \equiv (\bar{r}^g - c^{avg, s_i}(\bar{r}^g)) - (1 - \alpha^{s_i}) (\bar{r}^b - c_b(\bar{r}^b))$$

As before, the equilibrium is full pooling if $\phi^{s_i} > 0$, mixed if $-\frac{\pi}{1-\pi} (\bar{r}^g - c^{avg, s_i}(\bar{r}^g)) \leq \phi^{s_i} \leq 0$, fully bad-only if $\phi^{s_i} \leq -\frac{\pi}{1-\pi} (\bar{r}^g - c^{avg, s_i}(\bar{r}^g))$

Bank profits. For firms with signal s_i , bank profits are:

$$\Pi^{s_i}(r) = \begin{cases} [1 - \pi + \pi(1 - F^{s_i}(r))] (r - c^{avg, s_i}(r)) & \text{if } r \in [\underline{r}^{s_i}, \bar{r}^g] \\ (1 - \alpha^{s_i}) [1 - \pi + \pi(1 - F^{s_i}(r))] (r - c_b(r)) & \text{if } r \in (\bar{r}^g, \bar{r}^b] \end{cases}$$

The first region corresponds to rates that both types (weighted by α^{s_i}) would accept; the second region contains only bad types. After observing signal s_i , the bank's expected pooling cost of lending at rate r is $c^{avg, s_i}(r) \equiv \alpha^{s_i} c_g(r) + (1 - \alpha^{s_i}) c_b(r)$, where $c_\theta(r) = r_{CB}/p_\theta(r)$ as before.

Proposition C.1 (Signal Specific Equilibrium). *For $\pi \in (0, 1)$ and under signal acquisition, there exists a unique symmetric mixed-strategy equilibrium. Conditional on receiving signal s_i , banks randomize their rate offers according to the cumulative distribution function $F^{s_i}(r)$ that takes the same functional form as in the no-information case, with population share μ_b replaced by $(1 - \alpha^{s_i})$ and average cost $c^{avg}(r)$ replaced by $c^{avg, s_i}(r)$. The aggregate distribution is the mixture of signal-specific distributions weighted by the shares of the population receiving the signal σ_i .*

Explicitly, the signal-specific distributions are as follows, depending on the sign of ϕ^{s_i} :

Case 1: Pooling equilibrium ($\phi^{s_i} > 0$)

$$F^{s_i}(r) = \begin{cases} 0 & \text{if } r < \underline{r}^{s_i} \\ 1 - \frac{1 - \pi}{\pi} \left[\frac{\bar{r}^g - r}{r - c^{avg, s_i}(r)} \right] & \text{if } \underline{r}^{s_i} \leq r \leq \bar{r}^g \\ 1 & \text{if } r > \bar{r}^g \end{cases}$$

where $\underline{r}^{s_i}$ solves $F^{s_i}(\underline{r}^{s_i}) = 0$.

Case 2: Hybrid/bad-only equilibrium ($\phi^{s_i} \leq 0$)

$$F^{s_i}(r) = \begin{cases} 0 & \text{if } r < \underline{r}^{s_i,a} \\ F^{s_i,a}(r) & \text{if } \underline{r}^{s_i,a} \leq r \leq \bar{r}^g \\ F^{s_i,a}(\bar{r}^g) + F^{s_i,b}(r) & \text{if } \bar{r}^g < r \leq \bar{r}^b \\ 1 & \text{if } r > \bar{r}^b \end{cases}$$

where the equal-profit condition on each interval gives:

$$F^{s_i,a}(r) = \max \left\{ 1 - \frac{1 - \pi}{\pi} \left[\frac{(1 - \alpha^{s_i})(\bar{r}^b - c_b(\bar{r}^b))}{r - c^{avg,s_i}(r)} - 1 \right], 0 \right\}$$

$$F^{s_i,b}(r) = 1 - F^{s_i,a}(\bar{r}^g) - \frac{1 - \pi}{\pi} \left[\frac{\bar{r}^b - c_b(\bar{r}^b)}{r - c_b(r)} - 1 \right]$$

and the lower bounds $\underline{r}^{s_i,a}$, $\underline{r}^{s_i,b}$ are implicitly defined by $F^{s_i,a}(\underline{r}^{s_i,a}) = 0$ and $F^{s_i,b}(\underline{r}^{s_i,b}) = 0$.

Aggregate Rate Distribution. The distribution of rates across the entire population is the mixture of signal-specific distributions weighted by population fractions:

$$F(r) = \sigma_g F^{s_g}(r) + \sigma_b F^{s_b}(r)$$

Convergence to Perfect Information. As signal precision approaches perfection ($q_g, q_b \rightarrow 1$), posterior beliefs converge to certainty:

$$\alpha^{s_g} = \frac{q_g \mu_g}{q_g \mu_g + (1 - q_b) \mu_b} \rightarrow \frac{1 \cdot \mu_g}{1 \cdot \mu_g + 0 \cdot \mu_b} = 1$$

$$\alpha^{s_b} = \frac{(1 - q_g) \mu_g}{(1 - q_g) \mu_g + q_b \mu_b} \rightarrow \frac{0 \cdot \mu_g}{0 \cdot \mu_g + 1 \cdot \mu_b} = 0$$

where α^{s_i} denotes the probability of truly being a good type within the population receiving signal s_i , $i \in (g, b)$. This implies:

- **Good signal group:** $c^{avg,s_g}(r) \rightarrow c_g(r)$, so

$$\phi^{s_g} \rightarrow \bar{r}^g - c_g(\bar{r}^g) > 0$$

The CDF $F^{s_g}(r)$ converges to $F_g(r)$, the good-type distribution under perfect information.

- **Bad signal group:** $c^{avg,s_b}(r) \rightarrow c_b(r)$, so

$$\phi^{s_b} \rightarrow (\bar{r}^g - c_b(\bar{r}^g)) - (\bar{r}^b - c_b(\bar{r}^b)) \leq 0$$

The CDF $F^{s_b}(r)$ converges to $F_b(r)$, the bad-type distribution under perfect information.

Furthermore, $\sigma_g \rightarrow \mu_g$ and $\sigma_b \rightarrow \mu_b$, so the aggregate distribution converges to:

$$F(r) \rightarrow \mu_g F_g(r) + \mu_b F_b(r)$$

which is the perfect-information equilibrium from Section 5.4.

D Numerical Identification Checks for the Calibration Exercise

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D.1 From the offered distribution to the observed distribution

The equilibrium distribution $F(r)$ characterized in Section 5 is each bank's own symmetric mixed-strategy *posting* rule: the distribution from which a bank draws the rate it quotes to a given firm. It is not, in general, the same object as the distribution of rates actually *paid* by borrowers, once search is taken into account. A captive firm (mass $1 - \pi$) samples a single bank and pays a raw draw from $F(r)$. A shopper (mass π) samples both banks – each an independent draw from the same F , by the symmetry of the equilibrium – and accepts the lower of the two quotes. Consequently, conditional on transacting with a given bank, a shopper's paid rate is not a raw draw from F : winning requires that the offer beat an independent competing draw from F , which is more likely at low r than at high r .

Define $w(r)$ as the probability that a firm quoted rate r by a given bank actually transacts with it. A captive always accepts (contributing $1 - \pi$); a shopper transacts with this bank only if the rival's draw exceeds r , an event of probability $1 - F(r)$ (contributing $\pi(1 - F(r))$). Hence

$$w(r) = (1 - \pi) \cdot 1 + \pi(1 - F(r)) = 1 - \pi F(r). \quad (15)$$

At the bottom of the support $F(r) = 0$ and $w(r) = 1$: the cheapest possible quote always wins. At the top of the support, $F(r) = 1$ and $w(r) = 1 - \pi$: only captives remain, as every shopper has by then found a strictly better rival quote.

Aggregating captives (who pay a raw draw from F) and shoppers (who pay the minimum of two independent draws from F , with CDF $1 - (1 - F(r))^2$) across both banks

gives the density of *realized*, originated rates,

$$g(r) \propto f(r) w(r) = f(r) (1 - \pi F(r)), \quad (16)$$

where $f = F'$. The normalizing constant is, remarkably, always $1 - \pi/2$ regardless of the equilibrium regime (pooling, hybrid, or bad-only) governing F : because $F(r)$ is Uniform(0, 1)-distributed whenever $r \sim F$ (the probability integral transform), $\mathbb{E}[F(r)] = 1/2$ identically, so $\int f(r) (1 - \pi F(r)) dr = 1 - \pi/2$. Two limiting cases confirm the construction: at $\pi = 0$, $g = f$ exactly (with no search, offered and realized rates coincide); at $\pi = 1$, $g(r) \propto f(r)(1 - F(r))$, the density of the minimum of two draws, since every firm then shops.

For any moment $\mathbb{E}[r^k]$, this reweighting can be written as a correction to the moment computed directly under F :

$$\int r^k f(r) w(r) dr = \underbrace{\int r^k f(r) dr}_{\text{moment under } F} - \pi \underbrace{\int r^k f(r) F(r) dr}_{\text{correction term}}, \quad (17)$$

and the correction term admits a closed form in every equilibrium regime of the model (full pooling, hybrid, and bad-only), obtained by symbolic integration and verified against direct numerical quadrature. Throughout the calibration exercise of this section, the three targeted moments (mean, variance, and the rate–default correlation) are computed under $g(r)$ – properly reweighted by $w(r)$ and renormalized by $1 - \pi/2$ – rather than under $F(r)$ directly, since it is $g(r)$, not $F(r)$, that corresponds to the distribution of rates actually observed in loan-level data.

Section 5.4 summarizes three complementary numerical checks used to assess whether the parameter vector recovered by my calibration procedure (Table 4) for each rating class is a well-isolated global optimum of the GMM objective, given that the model’s regime-switching structure precludes a closed-form identification proof. This appendix documents each check, and its complete findings, in detail.

D.2 Local identification: rank and condition number of the moment Jacobian

At the estimated parameter vector $\hat{\theta} = (\hat{\pi}, \hat{q}, \hat{r}_g)$ for each rating class, I compute the 3×3 Jacobian of the three matched moments (mean, variance, and rate–default correlation) with respect to the three parameters by centered finite differences, with a one-sided fallback near a parameter bound (e.g. when $\hat{q}$ is at or close to 1). I report both the raw Jacobian and an elasticity-scaled version – entry (i, j) rescaled by $\hat{\theta}_j$ divided by the observed value of moment i (or by 0.01 when the observed correlation is within 0.01 of zero, to avoid dividing by a near-zero target) – so that condition numbers are comparable

in units across moments of very different natural scale (correlations of order 10^{-2} against rates and variances of order 1). Table 6 reports, for the elasticity-scaled Jacobian at each rating class’s $\hat{\theta}$, the rank, the condition number (ratio of largest to smallest singular value), and the parameter with the largest loading on the singular vector associated with the smallest singular value – the “weakest identified direction.”

The Jacobian has full rank (3/3) in every one of the eight rating classes, ruling out exact local non-identification. Condition numbers, however, range from 3.9 (rating 7) to 1097.0 (rating 3+), and the weakest direction loads overwhelmingly on q in six of the eight classes; the exceptions are rating 5, where π dominates, and rating 5+, where the loading is split between q and $\bar{r}_g$. The pattern is most pronounced for rating classes whose observed rate–default correlation is close to zero (ratings 6 and 7, and to a lesser extent the safest classes): correlation is the moment most informative about screening precision q , so a target correlation close to zero leaves q comparatively unpinned by the data, even though the model remains exactly locally identified in the formal rank sense.

D.3 Global search: dense grid visualization of the parameter space

Local curvature alone cannot rule out a second, disconnected minimum elsewhere in the admissible parameter cube $\pi \in [0.05, 0.95]$, $q \in [0.5, 1.0]$, $\bar{r}_g \in [0.5 \bar{r}_{\text{mean}}, 0.98 \bar{r}_b]$. I complement the Jacobian check with a dense grid search (tens of thousands of parameter combinations per rating class) evaluating the GMM objective at every point, and visualize, for each rating class, the subset of the cube whose implied moments fall within a chosen multiple of the best-achieved objective, as an interactive three-dimensional scatter colored by objective value, with the reported estimate marked separately. In every rating class, this “acceptable fit” region forms a single connected neighborhood around the reported estimate rather than several disjoint clusters – visual evidence against multiple, unrelated local optima, and a small fraction of the cube in every case (between 0.02% and 0.49% of grid points). Consistent with the Jacobian findings above, the matched region is elongated primarily along the q dimension for the rating classes with high condition numbers (3+ through 4); $\bar{r}_g$ is comparatively tightly pinned in every rating class, while q remains comparatively loosely pinned even for ratings 6 and 7, whose low condition numbers reflect a well-conditioned, but locally flat, objective in the q direction – consistent with their near-zero observed correlation leaving screening precision the least informative parameter to match.

As a complementary exercise, I also compute, for each rating class, the three surfaces in $(\pi, q, \bar{r}_g)$ -space along which the model exactly matches one observed moment at a time (mean, variance, or correlation individually). With three free parameters, each single scalar equation generically defines a two-dimensional surface; two such surfaces crossing is generically a one-dimensional curve, and all three crossing is generically an

isolated point – the geometric counterpart of the Jacobian’s rank condition. This exercise reveals that for rating 3+, the observed variance itself lies outside the model’s achievable range anywhere in the admissible cube: the maximum variance the model can generate there, at any $(\pi, q, \bar{r}_g)$, is approximately 0.28, against an observed variance of 0.40. More generally, model-implied variance falls short of the data in every rating class (Table 5), so the three moment-matching surfaces do not intersect exactly anywhere in the admissible region. This is the numerical counterpart of the two-type-versus-continuum discussion in Section 5: with only two latent borrower types, the maximum rate dispersion the model can generate is mechanically bounded, and this bound binds most severely for the safest rating classes.

D.4 Cross-checking the global optimum across equilibrium regimes

Finally, I exploit the fact that the equilibrium’s regime-switching structure is fully characterized by two threshold conditions, ϕ_{good} and ϕ_{bad} – one for the pool of borrowers who receive a favorable screening signal, one for the pool who receive an unfavorable signal – each independently placing that pool in one of three regimes (full pooling, hybrid, or bad-only lending), for nine theoretically possible regime pairs. Rather than assume ex ante which pairs are empirically relevant, I tag every point of the parameter grid with its actually realized $(\phi_{\text{good}}, \phi_{\text{bad}})$ regime pair. Only a subset of the nine pairs is ever realized within the admissible parameter region for a given rating class – six of nine for ratings 3+ through 5+, and only three of nine (all with the good-signal pool fully pooling) for ratings 6 and 7 – and the realized pairs always respect a simple ordering in which the good-signal pool is never in a strictly “less pooled” regime than the bad-signal pool. This is itself a useful by-product of the exercise: the theoretically possible nine-way regime space collapses, empirically, to a much smaller and highly structured set.

For each rating class and each empirically realized regime pair, I re-solve the GMM objective with the search restricted to that regime pair, implemented by penalizing, within the local optimizer, any candidate parameter vector whose realized regime pair differs from the target. In every rating class, the best objective value achieved across *all* admissible regime pairs coincides with the parameter vector reported by my baseline, regime-unconstrained estimation (Table 4), confirming that the baseline optimizer converges to the true global optimum of the objective function rather than a locally optimal point within a single equilibrium regime.

D.5 An alternative specification: fixing π at an observable proxy for search intensity

As a further check on the calibration, I re-estimate q and $\bar{r}_g$ for each rating class holding π fixed at an external, model-free proxy for search intensity, rather than estimating it jointly: specifically, I set π equal to the share of firms in that rating class holding at least

two active banking relationships, averaged over the sample period – the same underlying statistic cited in Section 5 as broadly consistent with the freely estimated $\hat{\pi}$ (on average 60% of firms across classes maintain at least two bank relationships). With π fixed at this value and r_{CB} still fixed at 1, I re-estimate $(q, \bar{r}_g)$ by the same grid-search-plus-L-BFGS-B GMM procedure used for the baseline (Table 4).

Table D.7 reports the results. The externally measured π is lower, and more compressed across rating classes, than the freely estimated $\hat{\pi}$ (0.40–0.68 against 0.71–0.77). Fit deteriorates in all eight rating classes: the objective value rises from 0.41 to 0.62 at rating 3+, and from essentially zero to 0.47 at rating 7.

Where the deterioration falls depends on the class. In the five safest classes the variance error barely moves and the extra loss lands on the mean: the model overshoots the class mean rate by 21–33% here, against 5–16% in the baseline. In the three riskiest classes both moments worsen at once — the share of observed variance the model reproduces falls from 81%, 80% and 94% to 67%, 30% and 45%, and the mean overshoots by 33–42%. Because rate dispersion in this model scales with $\pi \times (\bar{r}_g - c_{\text{avg}})$, a lower π compresses the dispersion the model can generate for a given $\bar{r}_g$, and the optimizer compensates by raising $\bar{r}_g$, which lifts the mean.

The constraint binds hardest at rating 3+, the one class in which the observed variance lies outside the model’s achievable range anywhere in the admissible $(q, \bar{r}_g)$ box at this fixed π . There $\hat{q}$ falls to its lower bound of 0.5 — no screening at all — against 1.00 in the baseline: at a low π the model buys what dispersion it can from pooling rather than from separation. In the other seven classes $\hat{q}$ stays within 0.05 of the baseline. The observed mean is unreachable in the four safest classes (3+, 3, 4+ and 4).

I read this exercise as informative rather than as grounds to prefer one specification over the other. The two-bank-relationship share is plausibly a lower bound on effective search intensity: maintaining a second banking relationship is necessary but not obviously sufficient for the kind of active rate shopping that π represents in the model, and the freely estimated $\hat{\pi}$ being somewhat higher is consistent with that interpretation. The exercise does, however, sharpen the point made throughout this appendix: the amount of rate dispersion the two-type model can generate is highly sensitive to π , and matching the variance moment closely requires π at or above roughly the level the baseline estimation recovers.

Table D.7: Alternative Specification: π Fixed at the Share of Firms with At Least Two Bank Relationships

Rating	π (fixed)	$\hat{q}$	$\hat{\bar{r}}_g$	Obj. (fixed π)	Obj. (baseline)	Var. reachable?
3+	0.600	0.500	3.055	0.62	0.41	No
3	0.651	0.985	3.709	0.45	0.35	Yes
4+	0.651	0.986	4.609	0.21	0.15	Yes
4	0.675	1.000	5.080	0.25	0.19	Yes
5+	0.670	0.999	5.417	0.25	0.20	Yes
5	0.562	0.991	6.662	0.22	0.04	Yes
6	0.399	0.552	6.347	0.64	0.04	Yes
7	0.426	0.554	8.895	0.47	0.00	Yes

Notes: π is fixed at the share of firms in each rating class with at least two active bank relationships, averaged over the sample period; $\hat{q}$ and $\hat{\bar{r}}_g$ are then re-estimated with r_{CB} fixed at 1, as in the baseline. “Obj. (baseline)” reproduces the objective value from Table 4; both runs match the same moments, so the two columns are directly comparable. “Var. reachable?” indicates whether the observed variance falls inside the model’s achievable range anywhere in the admissible $(q, \bar{r}_g)$ box at this fixed, lower π .

E Robustness: counting defaults as bad types [Back to main ↑](#)

The baseline calibration of Section 5 labels a firm bad when its Banque de France rating deteriorates over the three years following origination. Firms that default outright are not counted as bad. This appendix repeats the exercise under the alternative definition, in which the bad group also contains firms that default. Everything else — the rate regression, the residuals, the targeted mean and variance, $\bar{r}_b$, and p_g — is held fixed, so the two runs differ only through the three quantities the reclassification touches: the share of bad types μ_b , their success probability p_b , and the rate–default correlation.

Table E.8 shows how much those three move. For ratings 3+ through 5+ the reclassification is almost immaterial: μ_b shifts by at most 0.06 and p_b not at all. In the three riskiest classes it is large: μ_b roughly quintuples at rating 6 and sextuples at rating 7, and p_b collapses, because a firm labelled bad on the strength of its own default has, by construction, a default probability of one.

The estimates are unchanged where the definition is. For ratings 3+ through 5+, which account for the bulk of the sample, the two runs agree almost exactly. Comparing Table E.9 with Table 4, across those five classes $\hat{\pi}$ differs by at most 0.001, $\hat{q}$ by at most 0.021, and the objective value by at most 0.005; the fitted means and variances in Table E.10 reproduce Table 5 to two digits. The estimate of competition, which is the paper’s main quantitative result, is therefore not an artefact of how the bad type is defined: $\hat{\pi}$ lies in $[0.74, 0.77]$ here against $[0.71, 0.77]$ in the baseline.

The three riskiest classes are where the definitions part, and where the alternative is least reliable. At rating 5, $\hat{q}$ falls from 0.94 to 0.76; at rating 6 it rises from 0.53 to 0.61. The fit deteriorates sharply: the objective rises from 0.04 to 0.24 at

Table E.8: Inputs that change under the alternative bad-type definition

Rating	μ_b		p_b		Rate–default corr.	
	Baseline	With default	Baseline	With default	Baseline	With default
3+	0.846	0.846	0.997	0.997	0.025	0.023
3	0.749	0.751	0.996	0.996	0.033	0.034
4+	0.739	0.745	0.992	0.992	0.032	0.022
4	0.671	0.691	0.981	0.981	0.048	0.037
5+	0.491	0.549	0.965	0.965	0.048	0.044
5	0.268	0.415	0.908	0.630	0.017	0.049
6	0.066	0.344	0.784	0.200	0.000	0.068
7	0.069	0.446	0.630	0.200	0.000	0.000

Notes: “Baseline” is the definition used in Section 5, in which bad types are firms whose rating deteriorates; “With default” adds firms that default. For ratings 6 and 7 the raw data give a default frequency of exactly one in the bad group, which would imply an infinite cost of lending $c_b = r_{CB}/p_b$; p_b is therefore floored at 0.2, so the reported values for those two classes are the floor rather than an estimate.

Table E.9: Search frictions and information by rating class, defaults counted as bad

Rating	$\hat{\pi}$	$\hat{q}$	$\hat{r}_g$	Info Contr. (%)	Obj. Value $Q(\theta)$
3+	0.772	0.988	3.461	4.58	0.41
3	0.759	1.000	3.703	4.76	0.35
4+	0.737	0.978	4.609	2.83	0.16
4	0.758	1.000	5.092	3.81	0.19
5+	0.750	1.000	5.416	2.99	0.19
5	0.737	0.757	6.662	1.01	0.06
6	0.767	0.613	7.057	12.79	0.24
7	0.754	0.526	9.350	0.37	0.20

Notes: Columns as in Table 4, which reports the same quantities under the baseline definition.

Table E.10: Model fit, defaults counted as bad

Rating	Mean		Variance		Correlation	
	Data	Model	Data	Model	Data	Model
3+	1.67	1.94	0.406	0.153	0.023	0.023
3	1.83	2.08	0.443	0.188	0.034	0.034
4+	2.40	2.53	0.553	0.334	0.022	0.022
4	2.37	2.63	0.730	0.426	0.037	0.037
5+	2.57	2.81	0.867	0.494	0.044	0.043
5	3.36	3.51	0.985	0.740	0.049	0.049
6	3.63	4.15	1.137	0.600	0.068	0.070
7	4.79	5.39	1.843	1.064	0.000	0.003

Notes: Moments defined as in Table 5.

rating 6 and from 0.00 to 0.20 at rating 7, the model mean overshoots the data by 14% and 13% against 1% and 0% in the baseline, and the share of observed variance the model reproduces falls from 80% and 94% to 53% and 58%.

Two features of the exercise explain this, and both argue for keeping the baseline definition in the text. First, the definition is partly circular in these classes: a firm counted as bad because it defaulted has a default probability of one by construction, so p_b measures the labelling rule rather than the risk of the bad type. Second, the floor imposed on p_b to keep c_b finite binds at exactly 0.2 for ratings 6 and 7, which fixes $c_b = r_{CB}/p_b = 5$ against an upper bound $\bar{r}_b$ of 7.2 and 9.5. That gap is wide enough for information to generate more dispersion than the data contain, which is what produces the 12.8% information contribution at rating 6 — the one figure in either run that exceeds the 5% ceiling reported in Section 5. Excluding rating 6, the largest information contribution here is 4.8%, in line with the baseline. The same floor also makes the implied cross-subsidy at rating 6 unusually large: bad types pay an expected rate of 4.2 against a cost of $c_b = 5$, so the bank runs a loss of 0.8 per bad-type loan, covered by the surplus it earns on the good types pooled with them. I therefore read the rating-6 figure as an artefact of the floor rather than as evidence that screening contributes more to dispersion than the baseline implies.

Table E.11: Redistribution from a perfect signal, defaults counted as bad: percent of the rating class's total surplus

Rating	Bad types	Banks	Good types
3+	-0.893	0.883	0.010
3	-0.881	0.868	0.012
4+	-0.817	0.795	0.020
4	-0.747	0.693	0.052
5+	-0.640	0.547	0.094
5	-1.901	0.430	1.471
6	-12.231	0.396	11.834
7	-9.450	0.506	8.942

Notes: change in each party's expected surplus when signal precision moves from $q = 0.5$ to $q = 1$, as a percentage of the class's maximum total surplus. Rows sum to zero: every class is in full pooling at $q = 0.5$, so no firm is rationed at either end and information redistributes without creating surplus.

Table E.12: Redistribution from a perfect signal, defaults counted as bad: percent of each party's own surplus

Rating	Bad types	Banks	Good types
3+	-1.65	2.43	0.10
3	-1.87	2.29	0.08
4+	-1.83	1.94	0.14
4	-1.73	1.81	0.28
5+	-1.89	1.39	0.35
5	-7.61	1.04	4.35
6	-55.09	1.06	29.28
7	-33.82	1.30	27.02

Notes: the same changes, each divided by that party's own surplus at $q = 0.5$ rather than by the class total. Rows do not sum. This is the denominator relevant to a bank weighing the cost of acquiring precision, and to a borrower judging the size of the repricing.